

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
KATHMANDU ENGINEERING COLLEGE
KALIMATI, KATHMANDU



Major Project Final Report
on

NeuroVision: MRI-Based Brain Tumor Analysis

[Code Number: CT 755]

BY

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TO

DEPARTMENT OF COMPUTER ENGINEERING
KATHMANDU, NEPAL

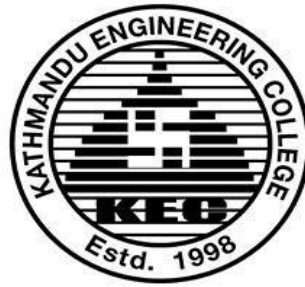
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NeuroVision: MRI-Based Brain Tumor Analysis

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PROJECT REPORT SUBMITTED TO
THE DEPARTMENT OF COMPUTER ENGINEERING
IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR
THE BACHELOR OF ENGINEERING



SUBMITTED BY

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ACKNOWLEDGEMENT

We would like to extend our thanks to the Department of Computer Engineering for giving us an opportunity to do this project. We would like to express our sincere gratitude to **Assoc. Prof. Shyam Dahal**, our Project Supervisor, for his constant supervision, dedicated mentorship and unwavering support throughout the entire duration of this project. His insightful guidance, technical expertise, and constructive feedback at every stage of the work played a crucial role in shaping the direction, quality, and depth of our project. His encouragement and timely suggestions helped us overcome challenges and maintain consistency and clarity in our work.

We would also like to thank **Assoc. Prof. Sudeep Shakya**, Head of Department of Computer Engineering, and **Assoc. Prof. Kunjan Amatya**, Deputy Head of Department of Computer Engineering, for providing us with the required guidance that we need for this project.

We are equally thankful to **Assoc. Prof. Sharad Chandra Joshi**, our Project Coordinator, for his invaluable guidance, effective coordination, and continuous support, which ensured the systematic planning and smooth progress of the project.

A special and heartfelt thanks is extended to **Mr. Nikhil Hari Dhungana**, Administrator of Annapurna Neuro Hospital, whose exceptional support during our field visit was invaluable to our project. His willingness to dedicate his time, patiently address our queries, and share in-depth clinical knowledge greatly enhanced our understanding of neurological practices and real-world applications relevant to our study. We are also deeply grateful to all the doctors, bioengineer, and staff members of Annapurna Neuro Hospital for their cooperation, insights, and encouragement, which provided us with practical exposure and enriched the overall outcome of our project.

Additionally, we would like to express our gratitude to our friends and family for their assistance and encouragement during the preparation of this report. Finally, our sincere gratitude goes to anyone who has helped us, directly or indirectly, in finalizing this report.

ABSTRACT

The brain is the most complex organ in the human body, serving as the central command center that regulates and coordinates all physiological, cognitive, and behavioral functions. Any abnormality in the brain can seriously affect a person's physical, mental, and emotional health. A brain tumor is an abnormal growth of cells in the brain, which can be either benign or malignant, and its impact depends on its size, type, and location. Early and accurate detection is very important for effective treatment and better patient outcomes. This project develops a deep learning based system for classifying brain tumors using MRI images, leveraging VGG16 as the backbone architecture through transfer learning to enhance feature extraction and classification performance. The dataset contains nearly 7,000 MRI images divided into four classes: glioma, meningioma, pituitary tumor, and no tumor. The training dataset includes 1,321 glioma, 1,339 meningioma, 1,595 no tumor, and 1,457 pituitary images, while the testing dataset includes 300 glioma, 306 meningioma, 405 no tumor, and 300 pituitary images. Additionally, 1,300 real-world hospital MRI images were used exclusively for external validation to assess the system's generalizability to unseen clinical data. The model classifies MRI images into the four categories. To improve interpretability, Grad-CAM is applied to visualize important regions in MRI images that influence the model's decisions, helping users understand how predictions are made. An NLP based chatbot, NeuroBot, is integrated into the system to provide user-friendly interaction, allowing users to ask questions, receive information about brain tumors, and perform similarity based queries. The system also supports automated report generation upon completion of predictions. The model achieved an overall classification accuracy of 98% with a peak F1-score of 0.98 on the test dataset, and 93.38% accuracy with an F1-score of 0.93 on the external hospital validation dataset, both evaluated over 20 epochs.

Keywords: Brain Tumor Detection, MRI Classification, Deep Learning, VGG16, Transfer Learning, Grad-CAM.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
BERT	Bidirectional Encoder Representations from Transformers
BraTS	Brain Tumor Segmentation
CAD	Computer-Aided Diagnosis
CNN	Convolutional Neural Network
CSS	Cascading Style Sheets
CT	Computed Tomography
DICOM	Digital Imaging and Communications in Medicine
FLAIR	Fluid Attenuated Inversion Recovery
GPU	Graphics Processing Unit
Grad-CAM	Gradient-weighted Class Activation Mapping
HTML	Hyper Text Markup Language
IARC	International Agency for Research on Cancer
IMA	Image File Format
LLM	Large Language Model
MRI	Magnetic Resonance Imaging
NLP	Natural Language Processing
RAG	Retrieval-Augmented Generation
ReLU	Rectified Linear Unit
SBERT	Sentence-Bidirectional Encoder Representations from Transformers
T1	T1-weighted Magnetic Resonance Image
T2	T2-weighted Magnetic Resonance Image
T1CE	T1-weighted Contrast Enhanced MRI
UI	User Interface
UML	Unified Modeling Language
VGG	Visual Geometry Group
VGG16	Visual Geometry Group 16-layer Network

CHAPTER 1: INTRODUCTION

1.1 Background Theory

The brain is the most complex and vital organ in the human body. It controls everything from breathing and movement to memory, emotions and decision-making. It is the center of thought, awareness and identity. Because of its role in every function of the body, any damage to the brain can lead to serious sometimes irreversible consequences. This makes brain-related diseases especially dangerous and emotionally overwhelming for both patients and their families.

According to medical research, the three most based common types of brain tumors: gliomas, meningiomas, and pituitary tumors are classified on their cellular origin and anatomical location. Gliomas originate from glial cells and are almost always malignant with glioblastoma, their most aggressive form, making up about 80 % of malignant brain tumors and showing a 5-year survival rate of just ~6.8 %. Meningiomas grow in the meninges and are mostly benign over 98 % are non-cancerous with a 5-year survival rate around 91 %, although ~2 % may become atypical or malignant. Pituitary adenomas form in the pituitary gland; the majority are benign, with roughly 0.1–0.2 % being malignant, yet even benign ones can cause serious hormonal imbalances and vision issues [1].

1.1.1 Gliomas

Gliomas originate from glial cells, which provide structural and metabolic support to neurons within the brain. These tumors typically occur within the brain parenchyma and are often aggressive and malignant in nature. Glioblastoma, the most severe form of glioma, accounts for a large proportion of malignant brain tumors and is associated with a poor prognosis and low survival rate.

Due to their infiltrative growth pattern, gliomas are difficult to distinguish from surrounding healthy tissue on MRI scans, making accurate detection and classification particularly challenging. Automated detection systems can play a vital role in identifying subtle features associated with gliomas at an early stage.

1.1.2 Meningiomas

Meningiomas arise from the meninges, the protective membranes that surround the brain and spinal cord. These tumors are generally slow-growing and are predominantly benign, with a high long-term survival rate. Although most meningiomas are non-cancerous, a small percentage can become atypical or malignant, potentially causing neurological symptoms due to compression of nearby brain structures. On MRI images, meningiomas often appear as well-defined masses, making them suitable candidates for image-based classification systems. Accurate identification of meningiomas is essential to differentiate them from more aggressive tumor types and guide appropriate clinical management.

1.1.3 Pituitary Tumors

Pituitary tumors, also known as pituitary adenomas, develop in the pituitary gland located at the base of the brain. While the majority of pituitary tumors are benign, they can significantly impact hormonal regulation and visual function due to their proximity to the optic nerves and endocrine system. Even small pituitary tumors may lead to serious clinical symptoms such as hormonal imbalance, vision impairment, and metabolic disorders. Detecting pituitary tumors through MRI imaging requires careful analysis of localized anatomical features, making them an important category for automated tumor classification systems.

Hence, brain tumors are one of the most serious and emotionally difficult health conditions. They can start in the brain itself or spread from other parts of the body like the breast or lungs. Early and accurate diagnosis is very important because it directly affects treatment success and survival. But identifying a brain tumor is not simple. Doctors often rely on MRI scans, where they must spot small changes in tissue by eye. This process takes time, can be tiring, and sometimes leads to mistakes, especially when tumors are small or look like healthy tissue.

In Nepal, this challenge is even bigger. Most hospitals outside major cities do not have access to advanced imaging tools or trained specialists. Many patients go undiagnosed or receive help only when the disease has already progressed. According to the IARC, more than 500 people were diagnosed with brain and central nervous system tumors

in Nepal in 2020. The real number may be higher since many cases go unreported. The lack of support tools like AI makes the workload heavier for the few doctors who are available [2].

This project focuses on building a smart system that can assist in identifying brain tumors from MRI scans. The aim is to explore how features like tumor size, shape, and texture can help detect if the tumor is harmful or not. While in the early stages, goal is to create something that supports doctors, saves time, and helps people in areas where medical help is hard to reach.

1.2 Problem Statement

Diagnosing brain tumors from MRI scans is one of the most demanding tasks in modern medicine. The process requires not only advanced technical knowledge but also a high level of concentration and clinical experience. Even skilled radiologists can find it difficult to detect subtle abnormalities, particularly when tumors are small, irregularly shaped, or visually similar to surrounding healthy tissue. The manual interpretation of MRI images is time-consuming and mentally exhausting, and under such conditions, the risk of human error becomes very real. When a diagnosis is delayed or incorrect, the consequences can be severe, ranging from inappropriate treatment and disease progression to the worst cases, loss of life.

This challenge becomes even more serious when viewed in the context of Nepal, where the healthcare system faces deep rooted structural limitations that go far beyond the diagnostic process itself. While cities have some level of medical infrastructure, the situation in rural and remote areas tells a very different story. Most district-level hospitals do not have access to MRI machines and even in facilities where basic imaging is available, the absence of trained radiologist means that the scans often cannot be properly interpreted. Patients living in mountainous or geographically isolated regions frequently have no choice but to travel for hours or even days to reach a hospital capable of handling their condition. By the time they arrive, the disease may have already progressed to a more advanced and difficult to treat stage.

Adding to this, Nepal does not yet have a centralized for tracking brain tumor cases at a national level. This makes it extremely difficult to understand the true scale of the

problem or to plan effective public health responses. In most clinical settings across the country, there are no AI-based or computer-aided tools to support the diagnostic process, which means that doctors must rely entirely on their own judgment under conditions that are often resource-constrained and high-pressure. This not only increases the likelihood of diagnostic errors but also places an unsustainable burden on an already stretched medical workforce. Low public awareness about the early warning signs of brain tumors further worsens the situation, as many patients seek medical help only when symptoms have become severe and treatment options have narrowed considerably.

The shortage of specialists, the lack of modern diagnostic infrastructure, the geographic barriers that separate patients from care and the complete absence of intelligent decision-support tools all point to the same urgent need. There is a genuine and pressing demand for an automated, accessible, and explainable system that can help detect and classify brain tumors from MRI scans, one that does not require a specialist to be physically present, can work effectively even in low-resource environments, and can provide doctors with reliable, interpretable support when they need it most.

1.3 Objectives

- To develop an explainable AI-based decision-support system for brain tumor classification from MRI with visual highlights.
- To develop an interactive chatbot for educational purposes and supportive clinical analysis.

1.4 Scopes and Applications

1.4.1 Scope:

This project is designed to work with existing hospitals and diagnostic systems and can be used in various medical institutions including hospitals, cancer centers, neurological clinics, and telemedicine platforms. It aims to assist radiologists and healthcare professionals by automatically detecting and classifying brain tumors from MRI scans using a deep learning model. The system takes MRI images as input and

outputs tumor predictions along with visual highlights, helping in faster and more accurate diagnosis. It can be installed in diagnostic labs, outpatient departments, rural health centers, and research facilities. By supporting early detection of brain tumors such as gliomas, meningiomas, and pituitary tumors, this project helps improve treatment outcomes, reduce diagnostic errors, and make healthcare more accessible and efficient.

1.4.2 Application:

- **Clinical Diagnosis Aid:** It helps radiologists and neurologists make faster, data-supported decisions in hospitals and diagnostic centers.
- **Rural Healthcare Support:** It assists in areas lacking specialized doctors by providing preliminary tumor detection from MRI scans.
- **Telemedicine Integration:** It enhances remote consultations by allowing doctors to view and analyze scan results through an AI-backed tool.
- **Educational Tool for Medical Training:** It provides a learning platform for medical students to understand brain tumor patterns and the role of AI in diagnostics.
- **Student Research and Projects:** It supports engineering and medical students in learning medical image processing, deep learning techniques, and AI-based healthcare applications through hands-on experimentation.
- **Academic Learning Platform:** It helps students understand the complete pipeline of data collection, preprocessing, model training, and evaluation in real-world medical AI systems.

CHAPTER 2: LITERATURE REVIEW

The research paper called “Review of deep learning: concepts, CNN architectures, challenges, applications, future directions” by Laith Alzubaidi, Jinglan Zhang and others states deep learning is a subset of machine learning that uses artificial neural networks with multiple layers to learn complex patterns directly from data. Unlike traditional machine learning approaches that rely on manually designed features, deep learning models automatically extract hierarchical features from raw inputs such as images. This property makes deep learning particularly effective for medical image analysis, where visual patterns are complex and often subtle [3].

The research paper titled “Detection of Brain Tumor – A Provided Method” by Dr. Samir Kumar Bandyopadhyay highlights the challenges of brain tumor detection using MRI due to variations in tumor shape, size, location, and intensity. The study emphasizes MRI as the preferred imaging modality because of its superior soft tissue contrast and non-invasive nature. It underscores the role of CAD systems in improving diagnostic accuracy and reducing manual effort. The paper reviews automated detection techniques using machine learning methods such as Artificial Neural Networks and Support Vector Machines. The provided approach applies MRI preprocessing and feature-based classification to identify malignant tumors, demonstrating that automated MRI-based systems can provide accurate and efficient tumor detection, although performance may vary with different datasets [4].

The research paper titled “Deep Learning and Machine Learning for Brain Tumor Detection: A Review, Challenges, and Future Directions” (2025) discusses the use of MRI for brain tumor analysis across axial, sagittal, and coronal planes, identifying axial views as the most commonly used in deep learning-based studies. The paper emphasizes the importance of MRI sequences such as T1, T2, and FLAIR, where T1 provides anatomical structure and tumor localization, T2 highlights edema, and FLAIR improves tumor visibility by suppressing cerebrospinal fluid signals. It categorizes datasets into public, clinical, and hybrid collections, noting that while public datasets often offer multi-modal MRI, clinical datasets are typically limited to single-sequence data. The study highlights that due to availability and consistency,

single-sequence axial MRI remains widely adopted in practical deep learning-based tumor classification systems [5].

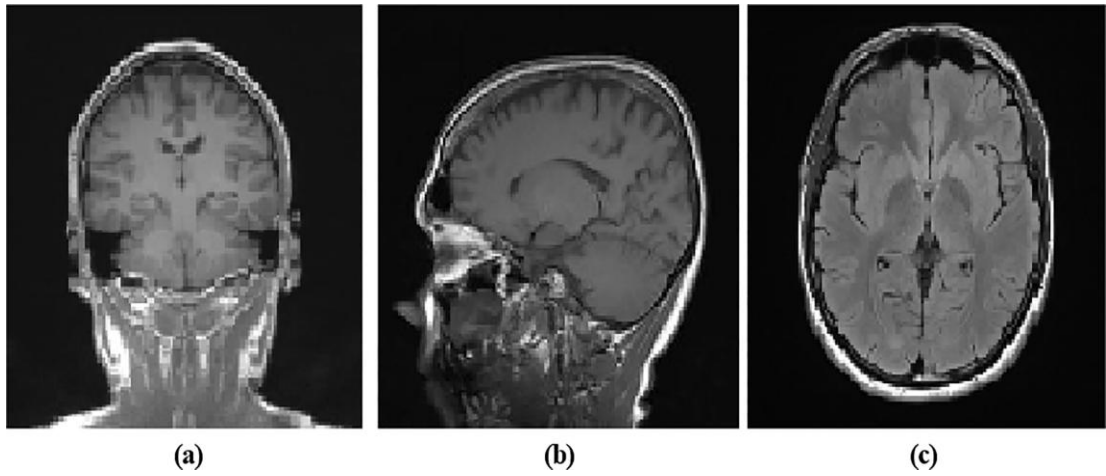


Figure 2.1: Fundamental MRI planes : (a) coronal (b) sagittal (c) axial

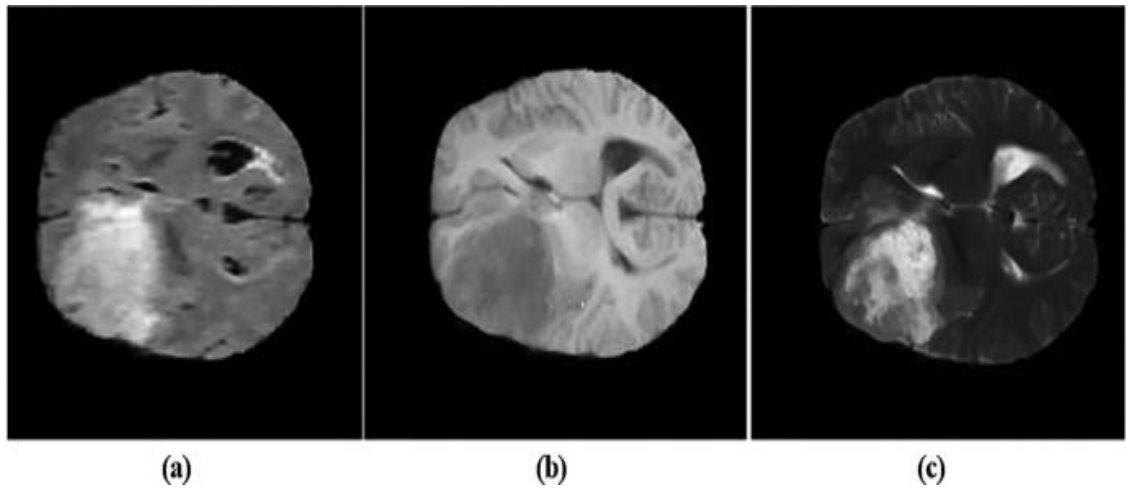


Figure 2.2: MRI brain tumor: (a) FLAIR image (b) T1 image (c) T2 image

The research paper titled “Modified VGG16 for Accurate Brain Tumor Detection in MRI Imagery” by Katuri Rama Krishna et al. [4] presents a deep learning approach for early detection of brain tumors using a modified VGG16 convolutional neural network. The study utilizes MRI images collected from Kaggle, applies preprocessing techniques to normalize and balance the dataset, and fine-tunes the VGG16 architecture by adding dropout and dense layers for binary classification. The provided model achieved an impressive accuracy of 99.08%, with high precision and recall,

demonstrating its effectiveness in accurately detecting brain tumors. The research emphasizes the simplicity and reliability of using VGG16 for computer-aided diagnosis, highlighting its potential to assist clinicians in timely and accurate diagnosis [6].

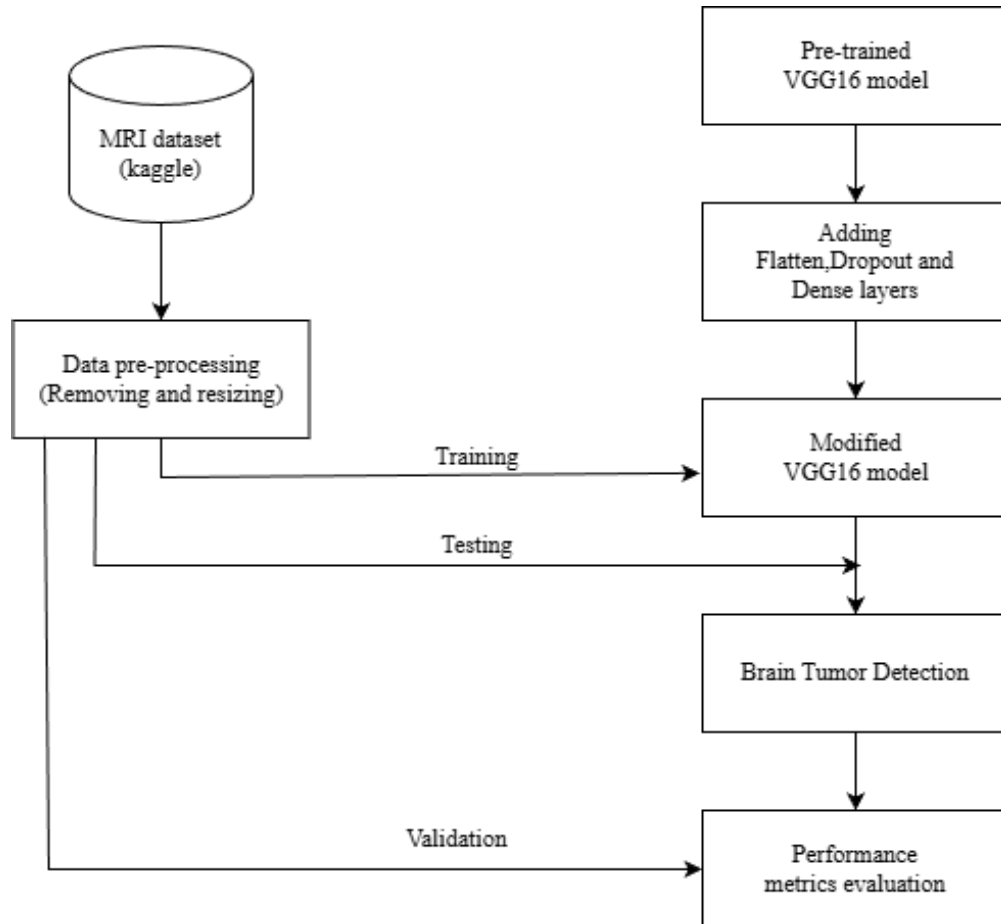


Figure 2.3 : Methodology for detecting brain tumor in MRI images

2.1 Deep Learning

Deep learning is a subset of machine learning inspired by the structure and function of the human brain. It uses artificial neural networks with multiple layers to process large and complex datasets, enabling machines to perform tasks such as classification, prediction, and decision making. It is widely applied in areas like medical imaging, where it can assist in detecting and analyzing abnormalities from raw data.

Feedforward networks involve passing input data through the network's layers, where each neuron computes outputs using weights, biases, and activation functions. This process continues sequentially until the final output is generated. The final output is represented as:

$$\text{Output} = \text{Weights} * \text{Inputs} + \text{Bias}.$$

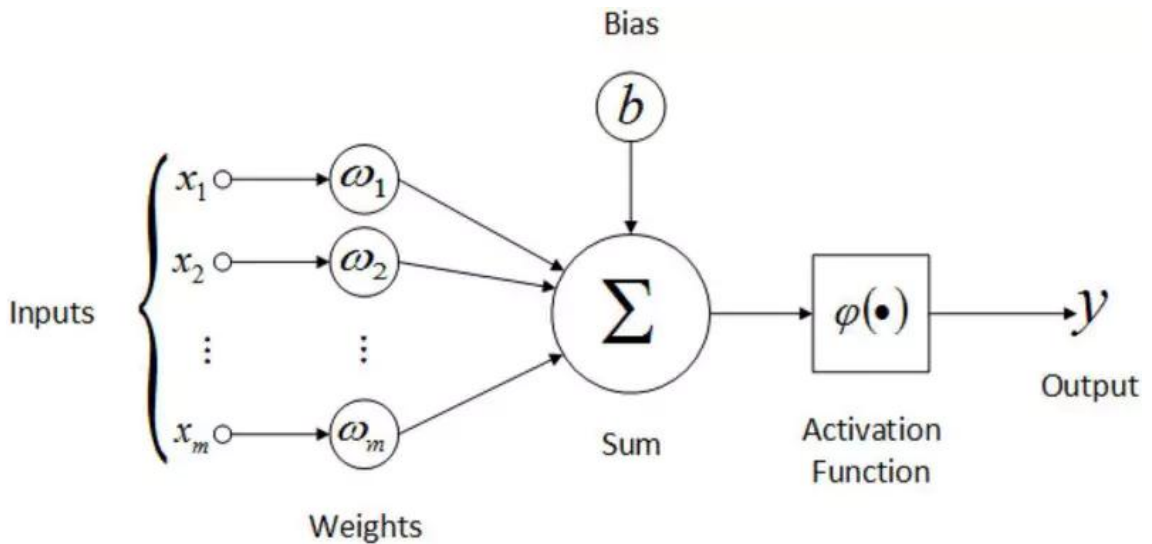


Figure 2.4: Feedforward Neural Network

Each neuron in a network receives inputs from the previous layer, and these inputs are multiplied by corresponding weights. The weighted inputs are summed together to form a single value. This value is then passed through an activation function, which determines whether the neuron activates or not. An activation function defines the output of a neuron by applying a mathematical transformation to its weighted sum. It introduces non-linearity, enabling the network to model complex relationships. Common activation functions include the sigmoid, hyperbolic tangent, and rectifier functions. Adjusting these weights during training helps the network learn to perform better.

Feedback networks, also called backpropagation, are used in training a neural network. It involves taking the error rate from feedforward propagation and feeding this loss backward through the neural network layers to fine-tune the weights.

2.2 CNN and Its Working

CNNs are a powerful deep learning architecture designed for tasks like image recognition and classification. They mimic how humans process visual information by learning features directly from the data. Here's how CNNs work:

2.2.1 Convolutional Layer

The convolutional layer is the heart of a CNN. It uses small, learnable filters, or kernels, that slide over the input data (such as an image) to perform a mathematical operation called convolution. This process extracts local features like edges, corners, and textures, creating feature maps. Each filter is designed to focus on specific patterns, and as the network deepens, it detects more complex features.

2.2.2 Pooling Layer

To reduce the spatial dimensions of the feature maps while retaining the most important information, CNNs use pooling layers. Max pooling is the most common type, where the largest value from a small region of the feature map is selected. This helps make the model computationally efficient, reduces overfitting, and ensures the network focuses on the most significant features.

2.2.3 Fully Connected Layer

After feature extraction and dimensionality reduction, the output is passed to one or more fully connected layers. These layers act like traditional neural networks, where all neurons are connected to every neuron in the previous layer. The fully connected layers interpret the high-level features and produce the final output, such as a classification label or a probability score.

Through these stages, CNNs process data hierarchically, learning patterns at multiple levels of abstraction. This makes them highly effective for computer vision tasks like object detection, facial recognition, and medical imaging.

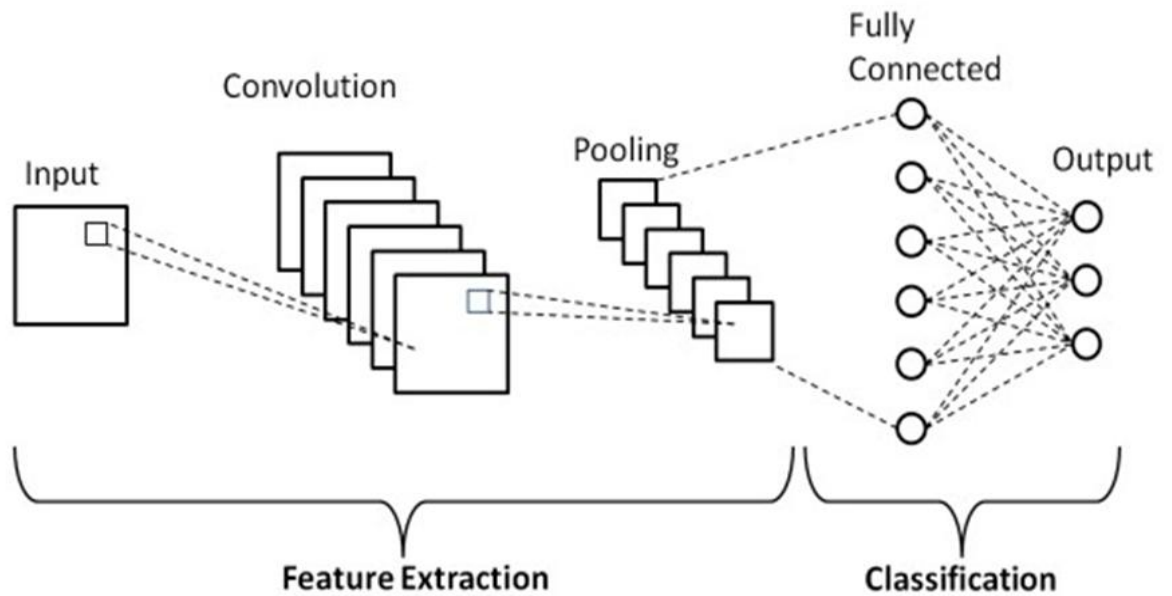


Figure 2.5: CNN Architecture

2.3 Transfer Learning

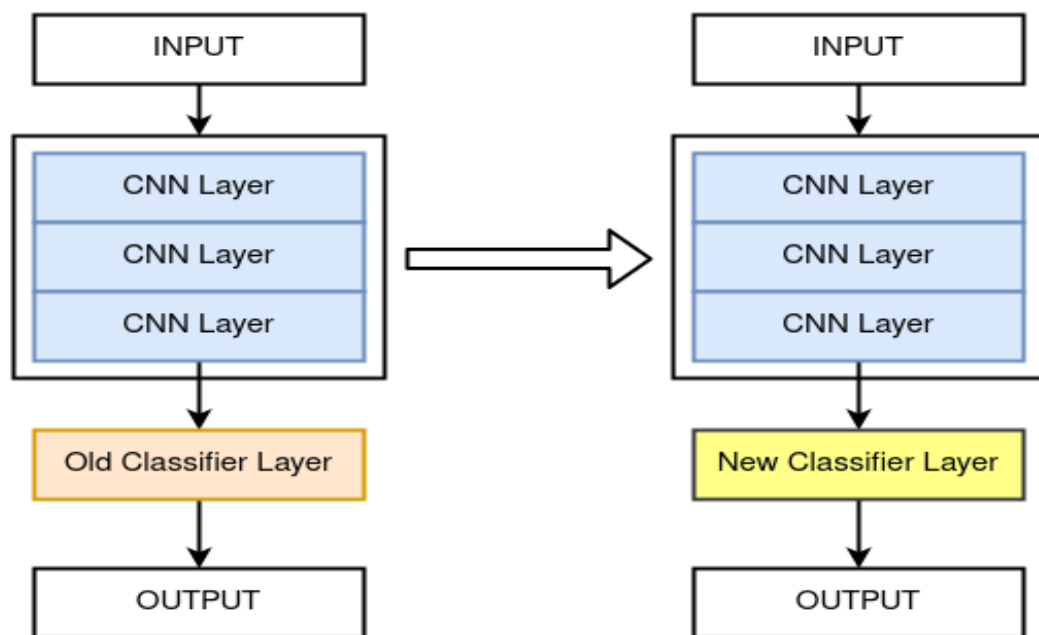


Figure 2.6 : Transfer Learning

Transfer learning is a machine learning approach where a model trained on one task is adapted for another related task. Instead of training a neural network from scratch, a pre-trained model is used as a starting point. This model has already learned useful

features from large datasets, which reduce training time and improves generalization on the new task, especially when labeled data is limited.

Common models used in transfer learning include VGG, ResNet, MobileNet, and EfficientNet. These models provide robust feature extraction and can be fine-tuned for tasks like image classification, object detection, or medical image analysis.

2.4 VGGNet

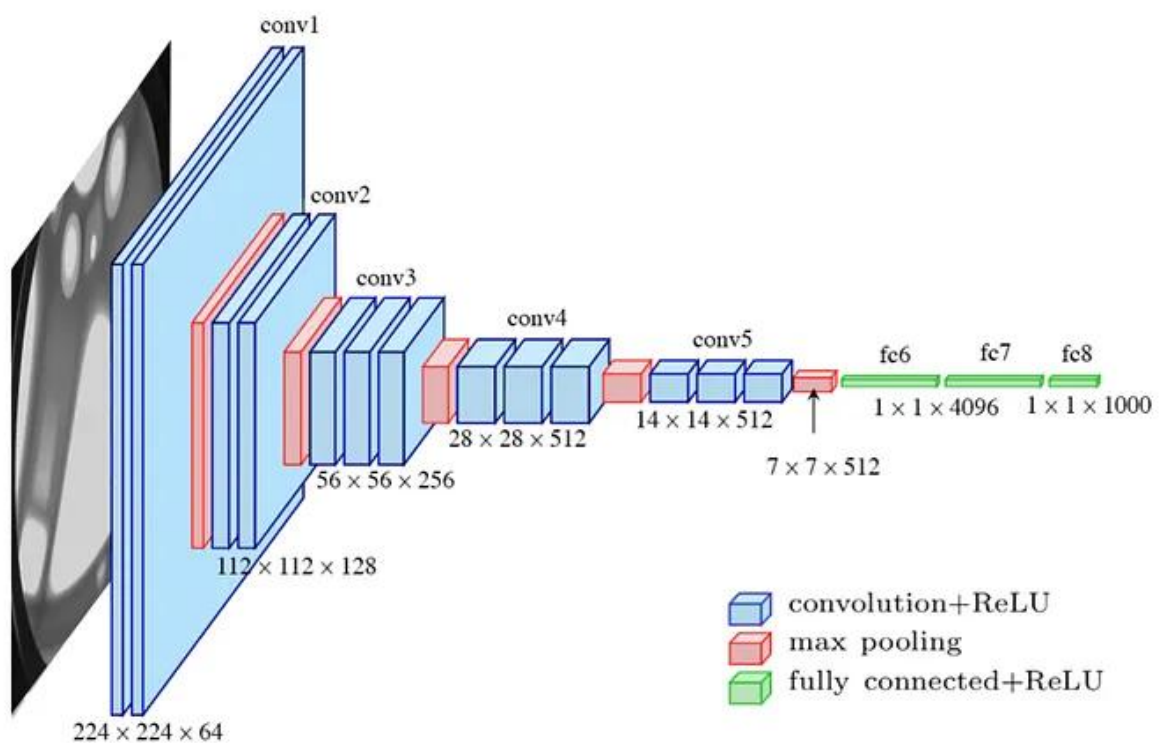


Figure 2.7: Architecture of VGGNet

VGGNet is a CNN architecture developed by the Visual Geometry Group at the University of Oxford. It is widely used in transfer learning due to its simple and uniform design, which relies on stacking small 3×3 convolutional filters throughout the network. The most popular configurations are VGG-16 (13 convolutional + 3 fully connected layers) and VGG-19 (16 + 3 layers), containing 138M and 144M parameters, respectively. These networks were historically important in achieving state-of-the-art results in ImageNet and other vision tasks.

The architecture is organized into five main blocks of convolutional and pooling layers. Each block contains multiple convolutional layers followed by a pooling layer. In the first block, the input passes through two convolutional layers (Conv1-1 and Conv1-2) and then a pooling layer. The second block follows a similar pattern with two convolutional layers (Conv2-1 and Conv2-2) and pooling. As the network goes deeper, the number of convolutional layers increases: the third, fourth, and fifth blocks each contain three convolutional layers followed by a pooling layer. This structured stacking of layers allows the network to progressively learn more complex features.

The convolutional layers use small 3×3 filters to scan the image and extract features. In the early stages, these layers detect simple patterns such as edges and textures. As the data moves deeper into the network, the layers begin to recognize more complex structures like shapes and object parts. At each stage, the spatial dimensions of the feature maps decrease while the depth (number of feature channels) increases, enabling the network to capture detailed and abstract representations of the input image.

After each block of convolutional layers, a pooling layer typically max pooling is applied. The pooling operation reduces the spatial size of the feature maps, which helps decrease computational complexity and prevents overfitting. It also retains the most important features by selecting the maximum values from small regions, ensuring that key information is preserved while redundant details are discarded. Once the feature extraction process is complete, the resulting feature maps are flattened into a one-dimensional vector and passed to the fully connected (dense) layers. VGG-16 contains three dense layers, where the first two perform high-level reasoning and feature combination, and the final dense layer produces the classification output. The last layer typically uses a softmax activation function to generate probabilities for each class, allowing the model to predict the most likely category for the input image.

2.5 Natural Language Processing

Natural Language Processing is a core discipline of Artificial Intelligence that focuses on enabling machines to process, analyze, and derive meaning from human language [13]. Human language exhibits characteristics such as ambiguity, polysemy, synonymy, and contextual dependence [14], which pose significant challenges for

computational modeling. NLP systems aim to bridge the gap between unstructured textual data and structured numerical representations suitable for machine learning algorithms.

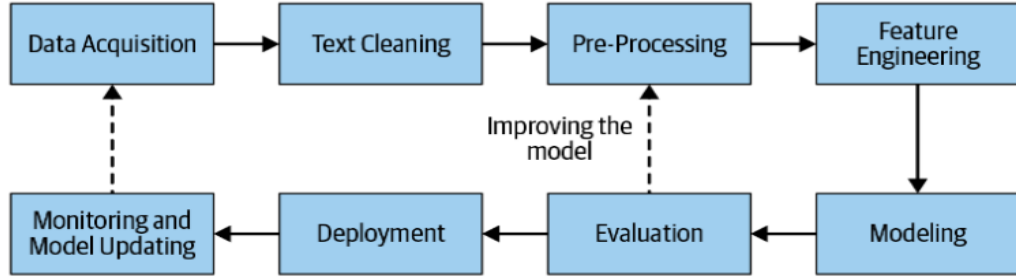


Figure 2.8 : NLP Pipeline

Formally, given a text sequence:

$$T = \{w_1, w_2, w_3, \dots, w_n\}$$

where w_i represents a token, NLP systems transform these tokens into numerical vectors using embedding functions:

$$f: w_i \rightarrow \{v\}_i \in \{R\}^d$$

This transformation enables linguistic processing through linear algebra, probability theory, and optimization methods.

2.5.1 Chatbots

Chatbots are software systems designed to simulate human-like conversation using natural language. Traditional chatbots relied heavily on rule-based logic, keyword matching, and decision trees [15]. Although effective for limited and predictable inputs, such systems failed to handle variations in language, complex sentence structures, or paraphrased queries.

With the advancement of NLP and deep learning, chatbots have evolved into intelligent systems capable of understanding contextual meaning. The provided chatbot leverages modern NLP techniques to interpret user intent rather than relying on exact keyword matches. By using semantic embeddings, the chatbot can respond appropriately even when the user's question is phrased differently from the stored medical information.

This significantly improves usability and makes the system more robust in real-world scenarios.

A chatbot query q is mapped to an embedding e_q , while each knowledge document d_i is represented as e_{d_i} . The chatbot selects the response maximizing semantic similarity:

$$d^* = \operatorname{argmax}_{d_i} \operatorname{cosine_similarity}(e_q, e_{d_i})$$

This formulation ensures deterministic and interpretable behavior. The NLP pipeline of the provided chatbot is structured into multiple stages that collectively enable accurate response retrieval. Initially, medical information related to brain tumors is collected and stored in textual form. This text is then preprocessed to remove inconsistencies such as unnecessary line breaks and formatting issues.

After preprocessing, the text is segmented into knowledge of chunks, where each chunk represents a complete and meaningful medical concept. Chunking ensures that responses are coherent and medically informative rather than fragmented. When a user submits a query, the input text undergoes the same processing pipeline to maintain consistency between stored data and incoming queries.

The chatbot is implemented as a retrieval-based NLP system, where user queries are matched with a predefined medical knowledge base using semantic similarity techniques. This design is particularly appropriate for healthcare-related applications, as it minimizes the risk of generating incorrect or misleading medical information and ensures that all responses are derived from validated sources.

2.5.2 Embedding-Based Semantic Retrieval

Embedding based semantic retrieval is a modern NLP approach that enables machines to retrieve relevant textual information based on semantic meaning rather than exact keyword matching [16]. In this paradigm, both user queries and knowledge base documents are transformed into dense numerical vector representations, known as embeddings, which capture contextual and semantic relationships within language.

Formally, let a user query be represented as a text sequence q , and a set of documents in the knowledge base be represented as $D = \{d_1, d_2, \dots, d_n\}$. An embedding function

$f(\cdot)$), implemented using a transformer-based encoder, maps each text sequence into a fixed-dimensional vector space:

$$e_q = f(q), \quad e_{d_i} = f(d_i), \quad e \in \mathbb{R}^d$$

where d denotes the dimensionality of the embedding space.

Once embedded, semantic similarity between the query and each document is computed using a similarity metric. In this project, cosine similarity is employed due to its effectiveness in high-dimensional vector spaces:

$$\text{sim}(e_q, e_{d_i}) = \frac{e_q \cdot e_{d_i}}{\|e_q\| \|e_{d_i}\|}$$

The most relevant document d^* is selected as:

$$d^* = \arg_max \text{sim}(e_q, e_{d_i}), \quad d_i \in D$$

This retrieval mechanism ensures that responses are selected based on semantic closeness rather than lexical overlap, allowing the system to handle paraphrased queries, synonyms, and variations in sentence structure.

Embedding-based semantic retrieval systems are particularly suitable for domain-specific and safety-critical applications, such as healthcare. Unlike generative language models, which may produce hallucinated or unverified responses, retrieval-based systems guarantee that all outputs originate from a predefined and validated knowledge base. This design choice significantly reduces the risk of misinformation and enhances the reliability of the chatbot. In the provided chatbot system, medical information related to brain tumors is preprocessed, segmented into semantically coherent chunks, and embedded using a Sentence-BERT model. During inference, user queries undergo identical preprocessing and embedding steps to ensure consistency. The system then retrieves the most relevant medical knowledge chunk based on semantic similarity and presents it as the chatbot response. This embedding-based semantic retrieval framework provides an effective balance between linguistic understanding, computational efficiency, and response reliability, making it well-suited for real-time medical information systems.

2.5.3 BERT

Bidirectional Encoder Representations from Transformers introduced deep bidirectional contextual learning. Unlike unidirectional models, BERT captures meaning based on both left and right context, producing contextual embeddings for each token. Given a sentence S , BERT generates contextual token embeddings:

$$H=\{h_1, h_2, h_3, \dots, h_n\}$$

However, sentence-level similarity using standard BERT requires pairwise inference, which is computationally inefficient for large document collections. Thus, an extension of the BERT architecture is SBERT, a specialized NLP model designed to generate meaningful embeddings for sentences and short text passages, optimized specifically for sentence-level semantic similarity tasks. Unlike standard BERT, which is computationally expensive for similarity comparisons, SBERT produces fixed-size embeddings that can be efficiently compared using similarity metrics.[17] SBERT enables semantic understanding by encoding entire sentences into vector representations that preserve contextual meaning. This makes it particularly suitable for tasks such as question-answering, information retrieval, and chatbot systems. In this project, SBERT is used to encode both user questions and knowledge base documents into embeddings. These embeddings are then compared using cosine similarity to identify the most relevant response. SBERT uses a Siamese or triplet network architecture, where identical transformer encoders process sentences independently.

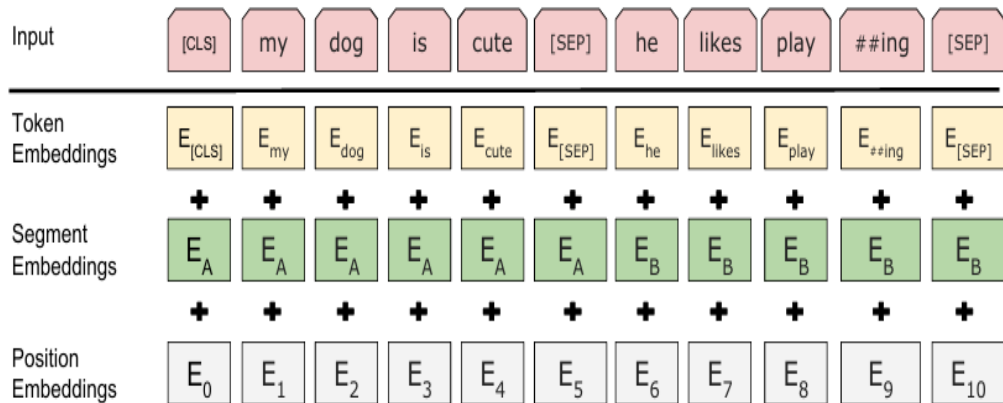


Figure 2.9 : BERT Input Representation

Each sentence is encoded as: $e_s = \text{Pooling}(H)$

Common pooling strategies include:

- Mean pooling
- Max pooling
- Classification Token pooling

Similarity between sentences is then computed efficiently:

$$\text{Similarity}(S_1, S_2) = \frac{e_{s1} \cdot e_{s2}}{\|e_{s1}\| \|e_{s2}\|}$$

This architecture reduces inference complexity from $O(n^2)$ to $O(n)$, making SBERT ideal for real-time chatbot systems [18].

The use of Sentence-BERT allows the chatbot to go beyond keyword-based matching and understand the intent and meaning behind user queries. This significantly improves the accuracy and relevance of responses, especially in a technical medical domain such as brain tumor information.

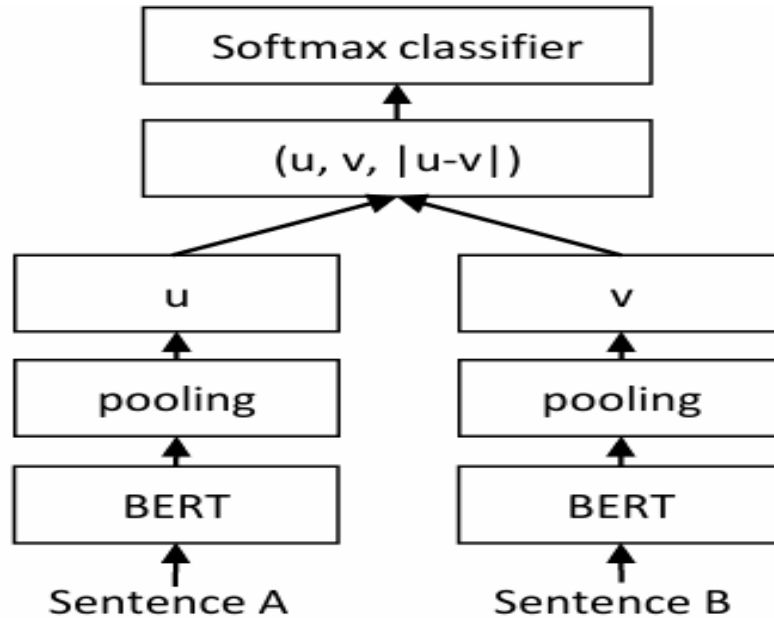


Figure 2.10 : SBERT Architecture With Classification Objective Function

2.5.4 LLM

LLMs are a class of deep learning models trained on massive amounts of text data to understand and generate human-like text. These models are built on the Transformer architecture, introduced by Vaswani et al. (2017), which uses a self-attention mechanism to capture long-range dependencies in text. LLMs are pre-trained on large corpora using unsupervised learning objectives such as next-token prediction or masked language modeling, enabling them to develop a broad understanding of language, facts, and reasoning [19] [20].

Some of the most prominent LLMs include OpenAI, LLaMA (Meta), PaLM (Google), and Gemini (Google DeepMind). These models contain billions of parameters and have demonstrated remarkable capabilities across a wide range of natural language processing tasks including question answering, summarization, translation, and dialogue generation.

In the context of conversational AI and chatbot development, LLMs play a critical role as the generative backbone of the system. Rather than relying on predefined rules or template-based responses, LLMs generate contextually relevant and fluent responses dynamically. When integrated into a RAG framework, the LLM does not rely solely on its parametric knowledge from training. Instead, it conditions its response on externally retrieved context, making the system more accurate, up-to-date, and domain-specific.

Gemini, developed by Google DeepMind, is a multimodal LLM capable of processing text, images, and other modalities. It offers a large context window of up to one million tokens, making it particularly suitable for RAG-based systems where multiple retrieved document chunks need to be injected into the prompt simultaneously. Its API accessibility, strong reasoning capability, and free tier availability make it a practical choice for research and academic projects involving intelligent chatbot development. The use of LLMs in chatbots represents a significant advancement over traditional approaches, enabling systems that can engage in multi-turn conversations, handle ambiguous queries, and provide grounded, coherent answers based on domain-specific knowledge bases [21] [22].

2.6 Grad-CAM

Grad-CAM is an explainability technique used to visually interpret the decisions made by CNN. Grad-CAM generates class-discriminative localization maps by computing the gradient of the class score with respect to the final convolutional layer's feature maps. These gradients are globally averaged to produce importance weights, which are then used to create a weighted combination of the feature maps, resulting in a heatmap that highlights the regions of the input image most influential in the model's prediction.

Formally, the importance weight α for each feature map k is computed as:

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}$$

where y^c is the score for class c , A^k is the k -th feature map, and Z is the total number of pixels.

The Grad-CAM heatmap L is then obtained as:

$$L_{Grad-CAM}^c = ReLU \left(\sum_k \alpha_k^c A^k \right)$$

The ReLU activation is applied to retain only the features that positively contribute to the target class, suppressing irrelevant regions.

In the context of medical image analysis, Grad-CAM is particularly valuable because it provides visual justification for model predictions, which is essential for building trust in AI assisted diagnostic systems. Rather than treating the model as a black box, Grad-CAM allows radiologists and medical professionals to visually verify whether the model is attending to clinically relevant regions such as tumor masses, rather than background artifacts or irrelevant tissue.

2.7 Existing System

2.7.1 VGG-16–Based Brain Tumor Detection System

Existing brain tumor detection systems based on the VGG-16 transfer learning model report high classification accuracies ranging from 96% on MRI datasets containing approximately 3,000–5,000 images. However, VGG-16 has a very large number of parameters, which increases computational cost and memory usage. Although it converges quickly with limited data and requires minimal parameter refinement, the

model lacks architectural efficiency and may not generalize optimally when the dataset size increases or when deployed in real-time clinical environments.

2.7.2 EfficientNet-Based Brain Tumor Detection System

EfficientNet-based systems have reported comparable or slightly higher peak accuracies than VGG-16 in some studies, typically after extensive fine-tuning, with performance improving mainly on datasets exceeding 5,000 MRI images. Despite its compound scaling and parameter efficiency, EfficientNet requires careful hyperparameter refinement and large training datasets to achieve stable results. On small or medium-sized datasets, its performance advantage is inconsistent, limiting its practical applicability for datasets commonly available in medical imaging.

2.7.3 ResUNet-Based Tumor Analysis System

ResUNet-based systems are primarily designed for tumor segmentation rather than classification, with reported Dice scores above 95% on large, pixel-annotated datasets such as BraTS. These systems require large-scale datasets with precise pixel-level tumor annotations, which are difficult and costly to obtain. Furthermore, the extensive parameter refinement and annotation dependency make ResUNet unsuitable for image-level brain tumor detection systems focused on classification accuracy.

2.7.4 Vision Transformer –Based Brain Tumor Detection System

Vision Transformer–based brain tumor detection systems report balanced classification accuracies in the range of 94% when trained on large or heavily augmented MRI datasets. However, Vision Transformer s contain a high number of trainable parameters and require substantial dataset sizes to generalize effectively. When trained on limited medical datasets, they show reduced stability, increased sensitivity to hyperparameter tuning, and lower generalization performance compared to CNN-based systems, making them less reliable for small-scale brain tumor detection applications.

2.8 Solutions Provided by the System

2.8.1 End-to-End Automated Classification Pipeline

The system delivers a fully automated workflow that processes brain MRI scans from input to final tumor classification. This streamlined pipeline reduces manual

intervention, minimizes diagnostic delays, and ensures consistent analysis across cases, making it suitable for real-time clinical deployment.

2.8.2 Deep CNN-Based Feature Extraction (VGG16)

A deep Convolutional Neural Network based on the VGG16 architecture is used to automatically learn complex and discriminative tumor features directly from MRI images. This enables precise identification of subtle patterns and variations in tumor structure that may not be easily visible through traditional analysis.

2.8.3 Transfer Learning for Limited Medical Data

The model leverages pretrained ImageNet weights through transfer learning, allowing it to achieve high performance even with limited medical imaging datasets. This approach enhances accuracy, reduces training time, and improves generalization across diverse MRI scans.

2.8.4 Consistent and Clinically Supportive Predictions

The system generates reliable and consistent predictions that support radiologists in clinical decision-making. By acting as a diagnostic assistance tool, it helps reduce interpretation variability and increases confidence in tumor classification results.

2.8.5 Retrieval-Based Medical Chatbot

An integrated retrieval-based chatbot provides accurate and verified answers to brain tumor-related queries. The chatbot uses a curated medical knowledge base, ensuring that responses are clinically relevant, trustworthy, and supportive for both medical professionals and patients.

2.8.6 Class-Discriminative Localization (Grad-CAM)

Grad-CAM visualization highlights tumor-relevant regions in MRI scans by showing which areas most influenced the model's decision. This improves clinical interpretability, allows visual validation of predictions, and strengthens trust in the AI system by making its reasoning more transparent.

CHAPTER 3: METHODOLOGY

3.1 Process Model

3.1.1 Iterative Waterfall Model

The Iterative Waterfall Model for the project, NeuroVision:MRI-Based Brain Tumor Analysis because it supports a structured yet flexible approach ideal for medical AI applications. This model breaks the project into well-defined phases starting from feasibility and requirements to design, coding, testing, and maintenance while still allowing feedback-driven improvements at each stage. Since the project involves multiple components such as MRI preprocessing, AI-based detection, explainability, and web deployment with chatbot support, the iterative nature of this model helps to refine and validate each part before moving forward. It also encourages detailed documentation and testing at every level, which is essential for building a reliable and trustworthy healthcare support system. Overall, the Iterative Waterfall Model helps us manage complexity, ensure accuracy, and deliver a system that meets real-world medical needs.

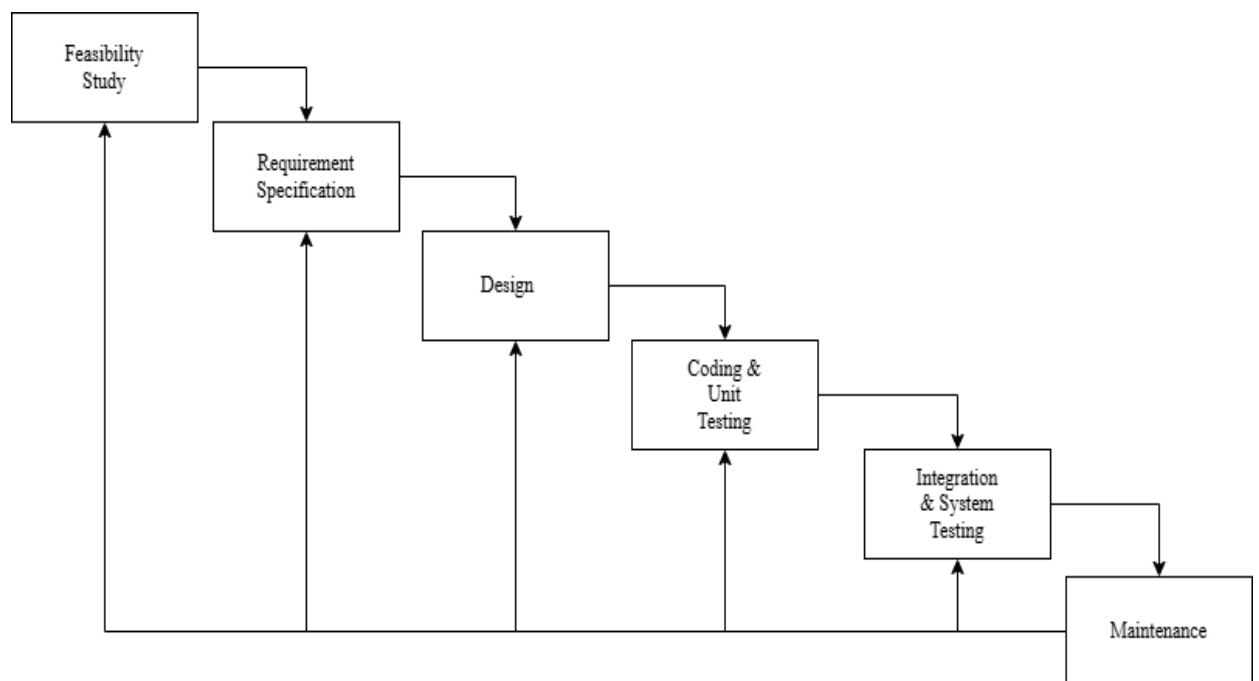


Figure 3.1 : Iterative Waterfall Model

The phases of Iterative Waterfall Model are:

i. Feasibility Study

The feasibility was assessed across three dimensions: technical, cost, and time. Technically, the project is achievable using open-source tools such as Python, Keras, Flask, and Google Colab, along with the publicly available Kaggle Brain MRI dataset. Cost-wise, the reliance on free platforms and tools makes it viable within a student project budget. In terms of time, development was planned in structured phases aligned with the academic timeline.

To support the initial analysis, semi-structured interviews were conducted with medical professionals at a Neuro hospital. It provided practical insights into real-world MRI workflows, tumor diagnosis procedures and the time radiologists typically spend analyzing scans.

ii. Requirement Specification

The system requires MRI image upload, tumor classification into four categories (glioma, meningioma, pituitary, and no tumor), and visual result display with heatmap output. The interface must be user-friendly, responsive, and accessible online. Core technologies selected include Keras, Flask, HTML/CSS, and Google Colab.

iii. Design

The system flow includes MRI upload, preprocessing, prediction with VGG16, and heatmap visualization using Grad-CAM. The UI will be kept dynamic with chatbot support and smooth frontend-backend integration.

iv. Coding and Unit Testing

Modules will be built for preprocessing, model training (using VGG16) and Flask integration. Each part must be unit-tested, and bugs to be fixed promptly.

v. Integration and System Testing

All modules should integrate and test with real MRI images. The system's accuracy, performance, and UI functionality (including the chatbot will be validated.

vi. Maintenance

The system will be monitored for performance and updated as needed. Future improvements include enhancing chatbot capabilities and adding 3d visualizations support.

3.2 System Block Diagram

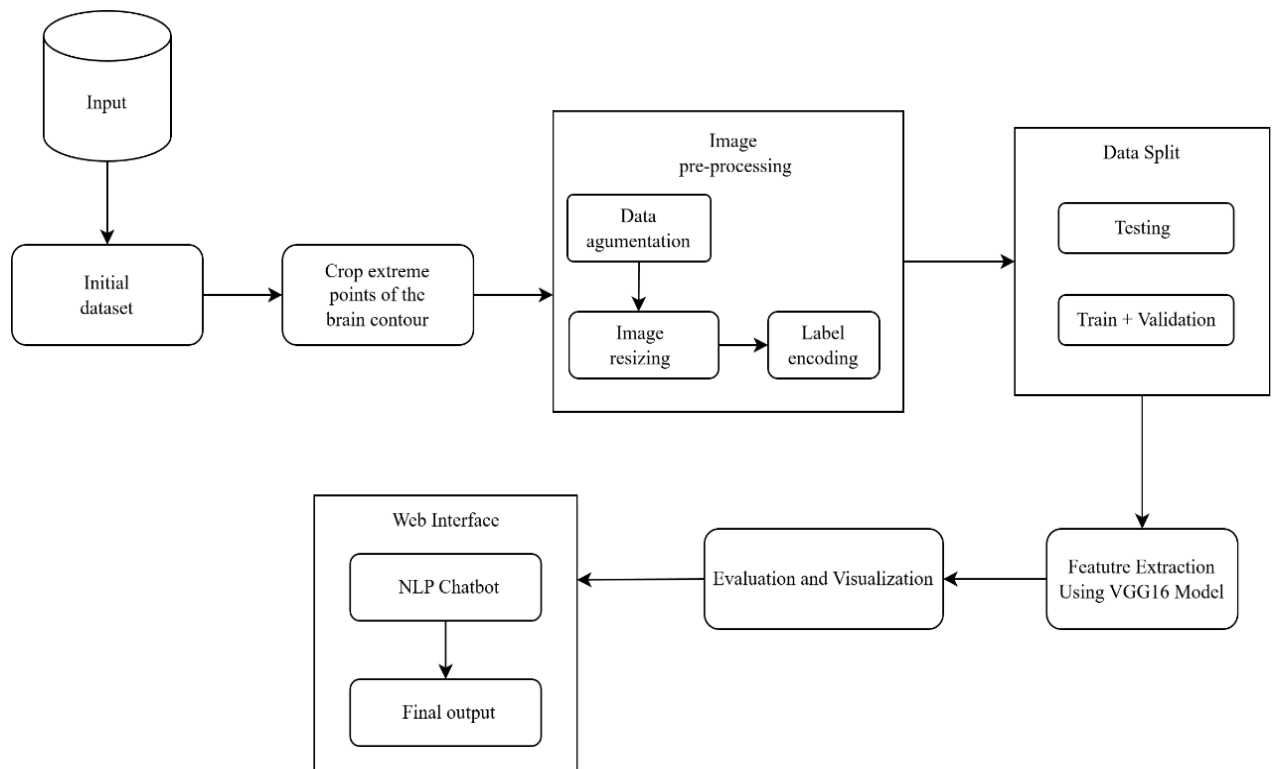


Figure 3.2 : System Block Diagram

- **Input Data Collection and Dataset Preparation:**

The brain tumor detection system begins with the input of brain MRI images collected from reliable medical datasets. These MRI images serve as raw data for training and testing the detection model. The collected images are organized into an initial dataset containing different classes such as tumor and non-tumor or multiple tumor types. At this stage, the dataset may include noise, unnecessary background regions, and variations in image size and quality. To improve model accuracy and focus on relevant information, the extreme points of the brain contour are cropped to remove unwanted areas such as skull edges and black borders. This process ensures

that only the essential brain region is retained, allowing the model to learn meaningful tumor-related features while avoiding distractions from irrelevant pixels.

- **Image Pre-Processing:**

Image pre-processing is applied to prepare MRI images for deep learning analysis. The images are augmented using techniques such as rotation, flipping, and scaling to increase dataset diversity and reduce overfitting. All images are resized to a fixed size compatible with the VGG16 model, ensuring uniformity across the dataset. Label encoding is also performed to convert class labels into numerical values so that the model can interpret them during training.

- **Data Splitting:**

After pre-processing, the dataset is divided into training, validation, and testing sets. The training and validation data are used to train and optimize the model, while the testing data is reserved for evaluating the model on unseen MRI images. This separation ensures fair performance evaluation and prevents data leakage.

- **Training mode:**

In the training phase, a pre-trained VGG16 deep learning model is loaded using transfer learning. The model layers are fine-tuned to adapt specifically for brain MRI images, allowing them to learn tumor-related features effectively. Hyperparameters such as learning rate, batch size, and number of epochs are optimized to improve training performance. After training, the final model is saved for future predictions and deployment.

- **Testing mode:**

The saved model is tested using unseen MRI images to evaluate its real-world performance. The model predicts whether a tumor is present, and its performance is measured using metrics such as accuracy, precision, recall, and F1-score. Grad-CAM visualization is applied to highlight the regions of the MRI image that influenced the model's prediction, improved interpretability, and trust in the system.

- **Web Interface and Final Output:**

A web-based interface is developed to make the system accessible and user-friendly. The interface includes an NLP-based chatbot that assists users by answering queries related to MRI analysis and tumor detection. The final output displayed to the user includes the tumor detection result, making the system practical for educational use.

3.3 Dataset Collection & Preprocessing

The primary dataset used for model training and testing was obtained from a publicly available brain tumor MRI dataset on Kaggle, consisting of nearly 7000 MRI images divided into four classes: glioma, meningioma, pituitary tumor, and no tumor. The training set contains 1321 glioma, 1339 meningioma, 1595 no-tumor, and 1457 pituitary images, while the testing set includes 300 glioma, 306 meningioma, 405 no-tumor, and 300 pituitary images.

Additionally, real-world MRI scans were collected from Annapurna Neuro Hospital in DICOM/IMA format, organized at the patient level with multiple slices per study. Due to strict data retention policies where scans are permanently deleted after two weeks, the number of available hospital images was limited. As a result, this hospital dataset was not used for model training but was instead used exclusively for real-world validation to assess how well the model performs on clinical data. For this project, T1-weighted, T2-weighted, T1CE, and FLAIR MRI sequences which provide complementary anatomical and pathological information essential for accurate brain tumor analysis.

For the Kaggle dataset, preprocessing included cropping to remove unnecessary background regions, resizing images to the fixed resolution required by the model, and intensity normalization to ensure consistent contrast. Data augmentation techniques such as random brightness and contrast adjustments were applied to improve generalization. Images were normalized to a $[0, 1]$ range and fed into the model using a custom batch data generator that also handled class label encoding. MRI scans consist of a stack of slices captured from the top to the bottom of the head. Since outer slices often contain non-brain regions such as the skull, eyes, and neck, these slices were

removed during preprocessing. Further preprocessing included cropping to remove unnecessary background areas, resizing images to a fixed resolution required by the model, and intensity normalization to ensure consistent contrast across all images. After preprocessing, the dataset was systematically organized into training and testing sets. The same preprocessing script is also used during testing to maintain consistency in data handling.

For the hospital dataset, DICOM/IMA files were read using the pydicom library and converted into JPEG (Joint Photographic Experts Group) format for compatibility. Hence, there are about 1300 images that includes 341 glioma, 301 meningioma, 347 no tumor, and 325 pituitary images. Min-max scaling was applied to normalize pixel intensity values into an 8-bit grayscale range, and outer slices containing non-brain regions were removed. No augmentation was applied to this dataset as it was used solely for validation purposes.

3.4 Training

Training the brain tumor detection model involves a series of systematic steps focused on utilizing pre-trained VGG16 weights through transfer learning and subsequently fine-tuning the network to accurately classify brain MRI images into tumor-related categories. Transfer learning is a crucial technique in medical image analysis, particularly when working with limited and specialized datasets. By initializing the VGG16 model with ImageNet pre-trained weights, the network benefits from previously learned visual features such as edges, textures, and spatial patterns. This significantly accelerates the convergence process and improves classification performance. Such an approach is highly effective for brain tumor detection, as it enables the model to transfer general visual knowledge to the domain-specific task of MRI image classification.

To initiate training, the MRI dataset is carefully preprocessed and labeled, ensuring that each image is accurately classified according to the presence and type of brain tumor. The VGG16 architecture, known for its deep convolutional layers, efficiently extracts hierarchical features from input images. To preserve these learned features, the convolutional layers of the base VGG16 model are frozen, and custom fully

connected layers are added to adapt the network to the classification task. The major parameters used during model training are:

- `--img 224`: Resizes input images to 224×224 pixels.
- `--batch 20`: Processes 20 images per batch for efficient training.
- `--epochs 20`: Runs training for 20 iterations over the dataset.
- `--weights imagenet`: Initializes the VGG16 base model with ImageNet pre-trained weights to leverage transfer learning and improve convergence.
- `--optimizer Adam (lr = 1e-4)`: Uses the Adam (Adaptive Moment Estimation) optimizer with a learning rate of 0.0001 for efficient and stable training.

During the training and fine-tuning phase, hyperparameters such as learning rate and dropout rate are carefully selected to improve generalization and reduce overfitting. Model performance is continuously evaluated using metrics such as training and validation accuracy, allowing for effective monitoring of learning behavior. Upon completion of training, the model architecture and learned weights are saved separately to facilitate future inference, retraining, or deployment of the brain tumor detection system.

3.5 VGG-16

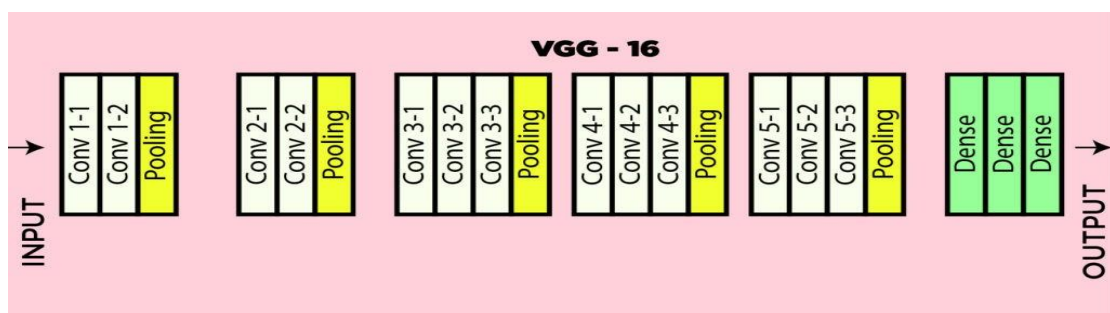


Figure 3.3 : Architecture of VGG-16

For detection, the saved model is reconstructed by loading the model.json file, which restores the complete architecture of the VGG16-based network, and the model.h5 file,

which loads all the trained weights and parameters. Once the model is ready, a new MRI image undergoes preprocessing steps such as resizing to 224×224 pixels, normalization, and reshaping to match the required input format. The preprocessed image is then passed through the VGG16 convolutional layers, where multiple hierarchical filters extract important visual features. In the early layers, the model identifies simple patterns like edges and textures, while deeper layers capture more complex structures such as tumor boundaries, shapes, and abnormal tissue patterns. This multi-level feature extraction allows VGG16 to effectively understand the MRI image at different levels of depth and detail.

After the convolutional base processes the image, the resulting feature maps are fed into the custom classification layers that were added on top of VGG16. The GPooling layer condenses the feature maps into a compact vector, which is then processed by a Dense layer to interpret the extracted features. The final Softmax layer generates probability scores for each tumor class such as glioma, meningioma, or pituitary tumor. These probabilities indicate how likely the input MRI belongs to each category. The class with the highest probability is automatically selected as the final prediction.

This detection workflow enables fast and accurate identification of brain tumors by combining VGG16's powerful feature extraction with a specialized classification mechanism. The model efficiently analyzes the MRI image, interprets the learned features, and provides a reliable prediction, making it highly suitable for automated medical image diagnosis.

3.6 Grad-CAM Integration

The integration of Grad-CAM into the system follows a structured pipeline designed to enhance the interpretability of the deep learning model. Grad-CAM is applied during the inference phase to provide visual explanations for the model's predictions. After an MRI image is input into the system and classified by the fine-tuned VGG16 model, Grad-CAM computes the gradients of the predicted class tumor score with respect to the feature maps of the final convolutional layers. These gradients are used to determine the importance of each feature map, which are then combined to generate a heatmap. The resulting heatmap highlights the regions of the image that contributed most significantly to the model's decision.

This heatmap is then superimposed onto the original MRI scan, producing a visual overlay that clearly indicates the areas of attention within the brain. In brain tumor classification tasks, such visualizations are particularly valuable, as they often align closely with regions identified by radiologists as tumor-affected areas. This provides an additional layer of validation beyond standard accuracy metrics and helps ensure that the model is focusing on clinically relevant features rather than irrelevant background information.

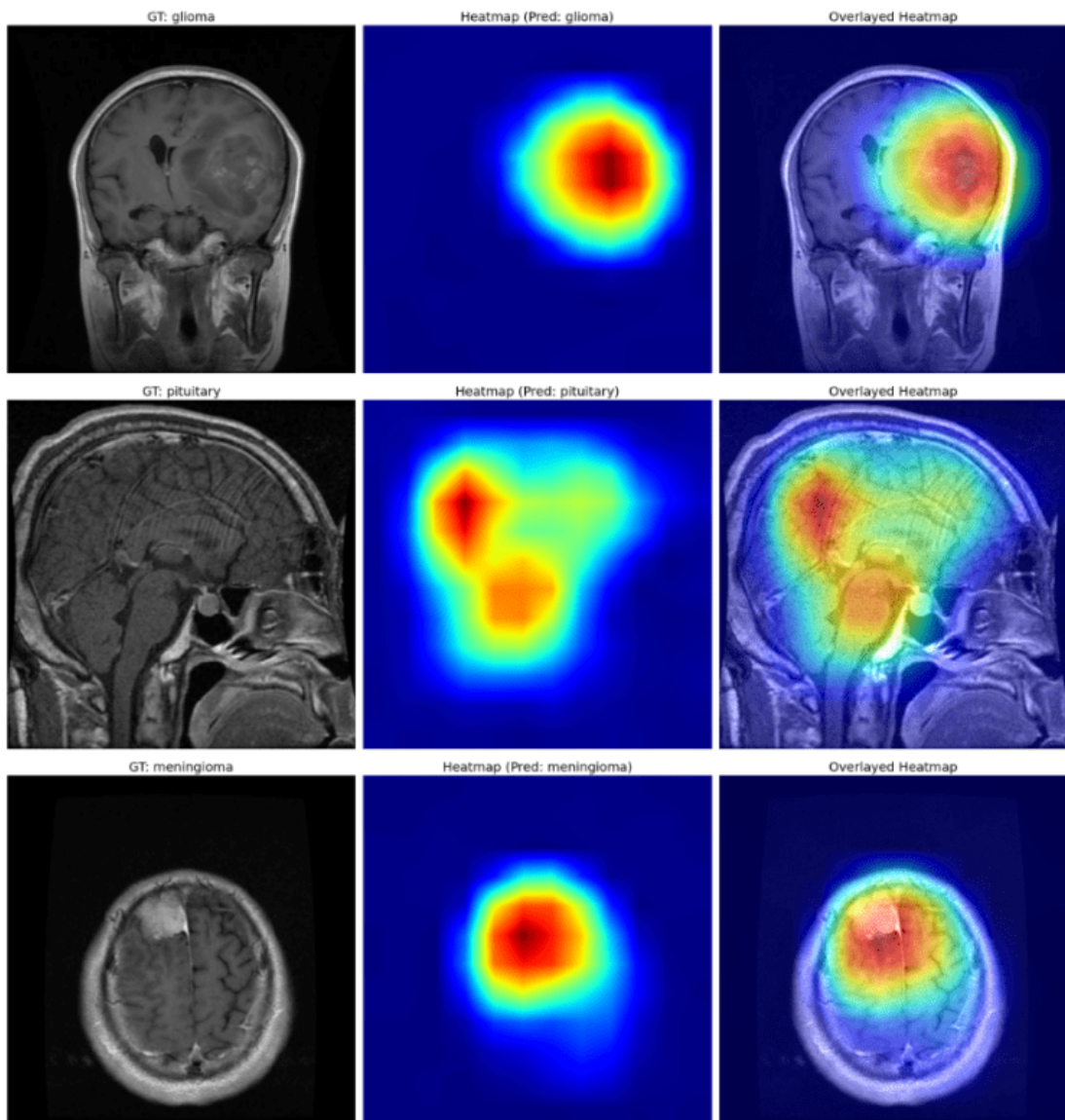


Figure 3.4 : Grad-Cam Visualization in Brain Tumor

3.7 RAG based Chatbot

In this project, the chatbot module of the brain tumor detection system is developed using a Retrieval-Augmented Generation approach. This method combines information retrieval from a structured knowledge base with response generation using a large language model. Such an approach is especially suitable for medical applications because it helps ensure that responses remain accurate, consistent, and based on verified medical information. This is important when dealing with sensitive health-related queries where reliability and clarity are essential.

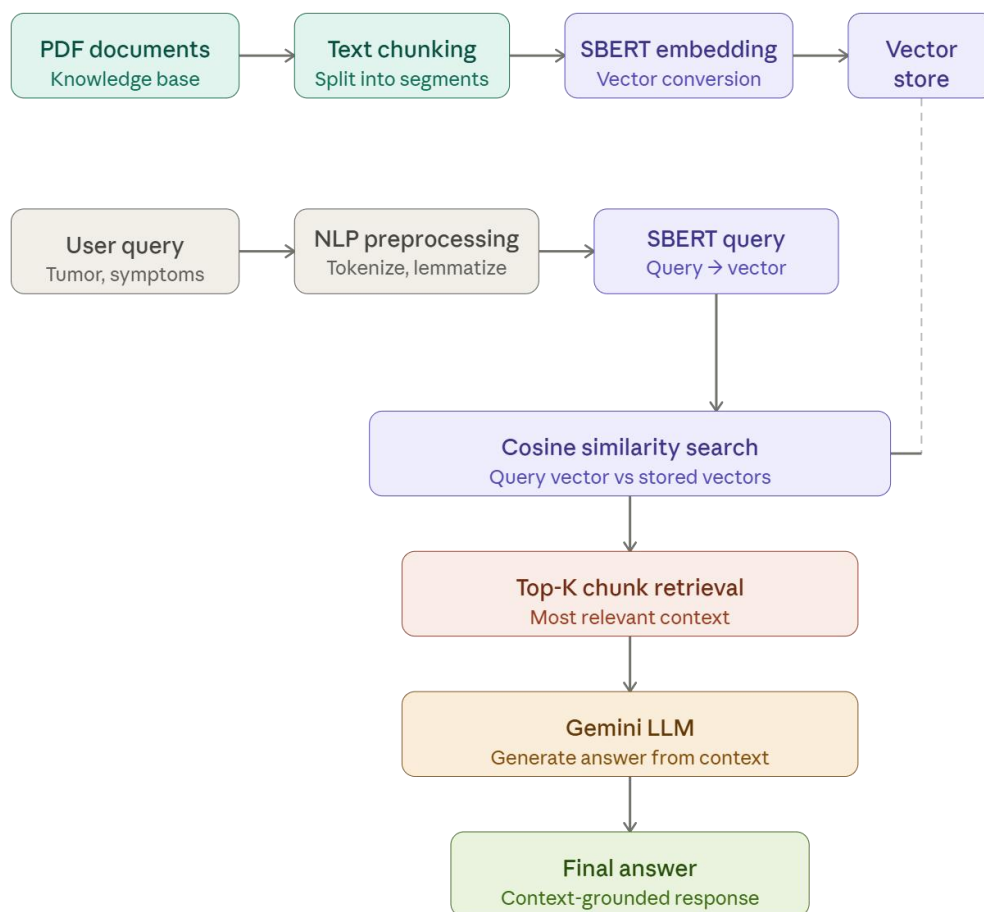


Figure 3.5 : RAG workflow in NeuroVision System

The process begins with creating a structured knowledge base that includes validated medical literature, brain tumor datasets, clinical guidelines, and relevant case studies. Each piece of information is converted into numerical vector representations using transformer-based embeddings. For this purpose, SBERT is used, as it captures the semantic meaning of text and converts it into high-dimensional vectors. These vectors

are then stored in a vector database, which allows the system to quickly search and retrieve relevant information based on similarity.

When a user asks a question, the input is first processed using basic NLP techniques such as tokenization, stop-word removal, and entity recognition. This helps the system identify important medical terms like tumor type, symptoms, or prediction confidence. After preprocessing, the query is converted into a vector using the same SBERT model used during knowledge base creation. The system then calculates cosine similarity between the query vector and stored document vectors to find the most relevant information. The top-K matching results are retrieved and passed as context for response generation.

The retrieved information is then provided to Gemini, a large language model developed by Google DeepMind. Gemini generates responses based on the retrieved context instead of relying only on its pre-trained knowledge. This reduces the possibility of generating incorrect or misleading information, which is especially important in healthcare-related systems. Additionally, ranking and threshold mechanisms can be applied so that only highly relevant information is used for generating responses, further improving reliability.

By combining SBERT-based semantic retrieval with Gemini-based response generation, the chatbot can better understand medical queries, retrieve relevant knowledge efficiently, and provide clear, context-aware answers. This approach also improves the scalability of the system, allowing it to handle more queries without affecting accuracy. Overall, the RAG-based chatbot improves the usefulness, transparency, and reliability of the brain tumor detection system, making it helpful for educational, clinical, and decision-support purposes.

3.8 Flowchart

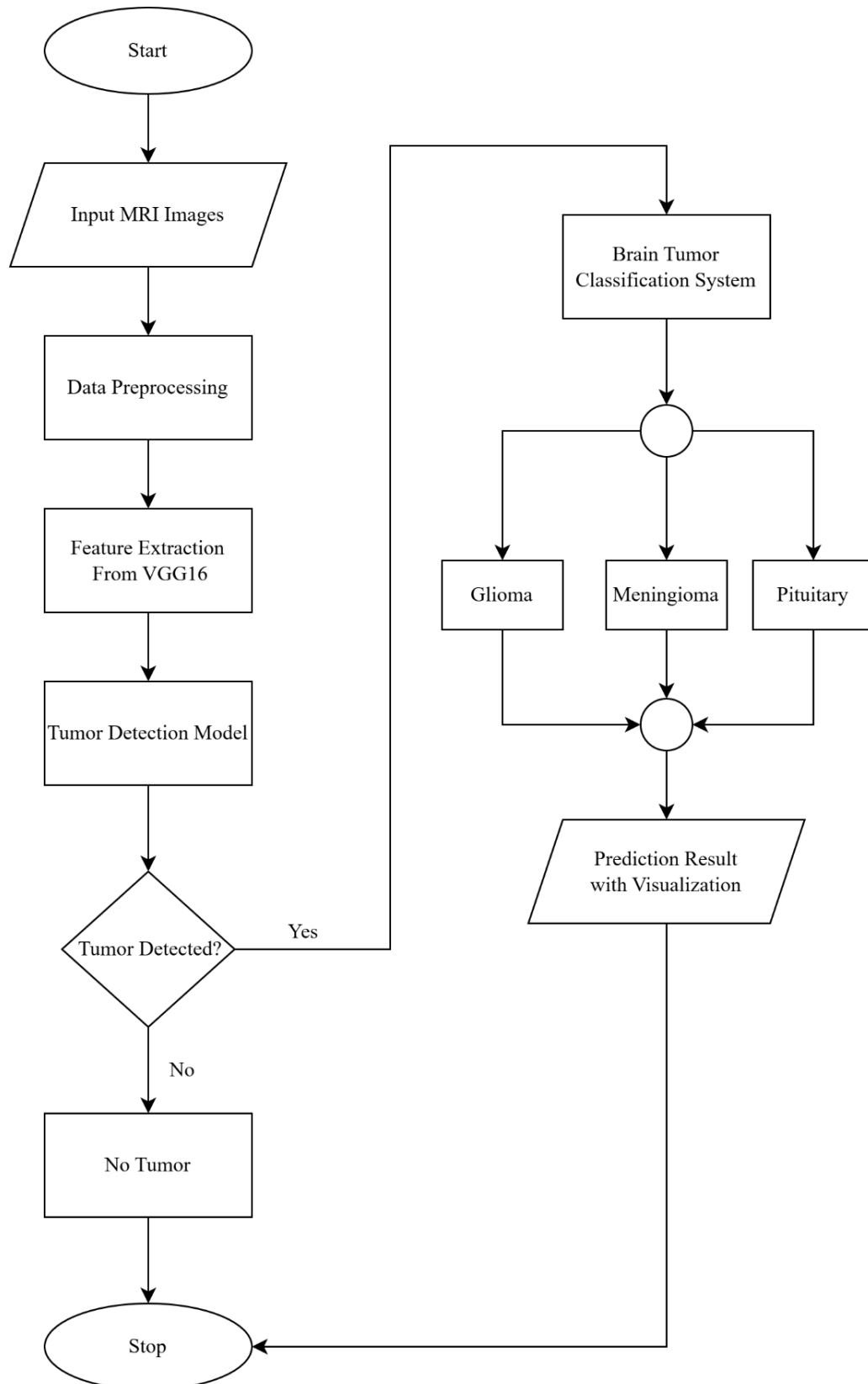


Figure 3.6 : Flowchart of System

3.9 UML Diagrams

3.9.1 Use Case Diagram

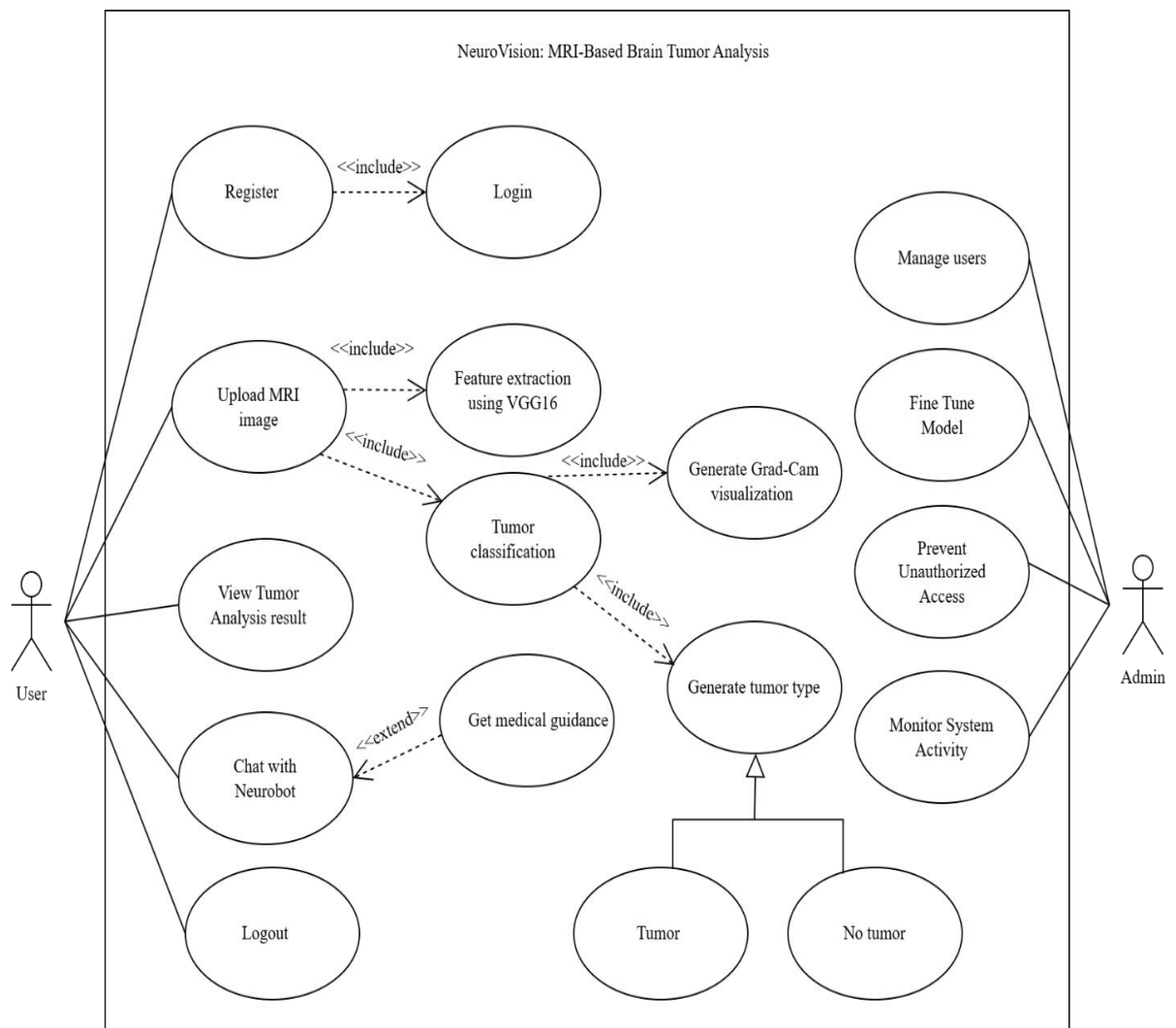


Figure 3.7 : Use Case Diagram

The use case diagram represents the functional interactions between external actors and the NeuroVision: MRI-Based Brain Tumor Analysis System. Two primary actors are identified: the User and the Admin.

The User interacts with the system to register, login, upload MRI images, view tumor analysis results, and consult the NLP-based chatbot for medical guidance. The diagram demonstrates that the login process includes optional extensions such as “Save

Password” and “Forget Password,” which provide additional usability features beyond the core login functionality.

Once the user uploads an MRI image, the system automatically performs internal processing through a series of machine learning operations. These include MRI preprocessing, feature extraction using the VGG16 deep learning model, and tumor classification. After classification, the system generates a tumor type and a corresponding confidence score.

These internal processes are depicted using include relationships, indicating that they are mandatory system operations triggered by the main user action. Following result generation, the user can view the predicted tumor type and confidence score, while the system may optionally store these predictions for performance tracking or future analysis.

The Admin actor oversees the operational integrity of the system. Administrative use cases include managing users, maintaining and updating datasets, training and updating the VGG16 model, and monitoring model performance metrics. Through these interactions, the Admin ensures that system accuracy, dataset quality, and model robustness are continuously improved and maintained.

Overall, the use case diagram provides a clear representation of how NeuroVision integrates user interaction, automated tumor detection, and administrative management to deliver a complete and efficient tumor analysis workflow.

3.9.2 Sequence Diagram

The sequence diagram illustrates the dynamic interaction between the user and the main components of the brain tumor detection system. The process begins when the User uploads an MRI brain image through the System interface. Upon receiving the input, the system performs necessary preprocessing operations, such as resizing and normalization, to prepare the image for analysis.

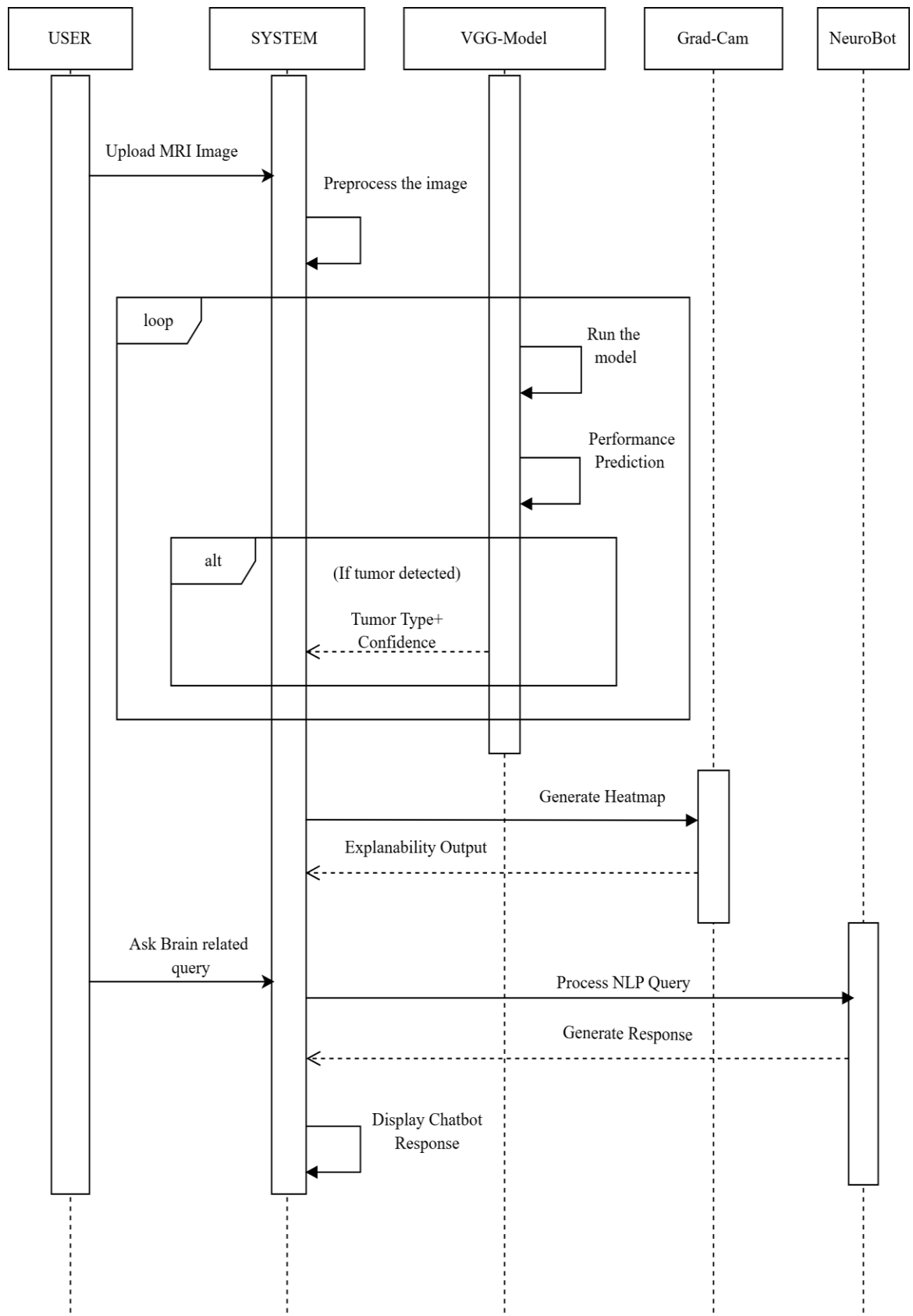


Figure 3.8 : Sequence Diagram

The loop fragment in the diagram represents internal repetitive operations involved in preparing and validating the input data. After preprocessing, the system forwards the image to the VGG-Model, which performs multi-class classification to identify the type of brain tumor present. The model analyzes the MRI image and returns the prediction results to the system. Activation bars in the diagram indicate the time during which each component is actively executing its respective tasks.

To enhance interpretability, the system then interacts with the Grad-CAM module, which generates visual heatmaps highlighting the regions of the MRI image that influenced the model's decision. These explainable outputs are returned to the system and displayed to the user along with the classification result.

Additionally, the diagram shows an alternative interaction flow involving NeuroBot, the NLP-based chatbot. In this flow, the user submits brain tumor related queries, which are processed by the chatbot to provide informative and educational responses. Overall, the sequence diagram demonstrates the integration of tumor detection, explainable AI, and NLP components into a unified and efficient workflow.

3.10 Tools Used

1. **Python:** Python is the core programming language used for model development, data preprocessing, training, evaluation, and deployment due to its simplicity and strong support for machine learning and image processing libraries.
2. **JavaScript:** JavaScript adds interactivity and dynamic behavior to web pages, allowing for features like form validation, animations, and real-time updates, enhancing user experiences and extending the functionality of web applications.
3. **HTML & CSS:** HTML and CSS are used to design and style the frontend of the application. HTML structures the content, including input fields for uploading MRI images and sections for displaying results. CSS enhances the appearance of the interface, ensuring it is visually appealing, responsive across devices, and user-friendly for non-technical users interacting with the system.

4. **Kaggle:** Kaggle is a comprehensive platform offering datasets, coding environments, and community support for data science and machine learning projects. Kaggle provide access to relevant datasets.
5. **Google Colab:** Google Colab is a free, cloud-based platform provided by Google that allows users to write and run Python code in a Jupyter Notebook environment. It provided access to free GPU resources, which significantly reduced training time and improved performance. Colab also made it easy to import datasets from sources like Kaggle and Google Drive, run experiments interactively, and share progress with team members during development.
6. **NumPy:** NumPy is the foundation for numerical computing in Python. It provides efficient arrays and mathematical functions for calculations needed in machine learning algorithms.
7. **Pandas:** This is a powerful library for data manipulation and analysis in Python. It allows us to work with data structures like Data Frames and Series, enabling tasks like data loading, cleaning, and transformation.
8. **Flask:** Flask is a lightweight web framework in Python used to connect the frontend to the backend model. It handled image uploads, routed requests, and returned prediction results to users in real-time.
9. **TensorFlow & Keras:** TensorFlow and Keras are deep learning libraries that are used to build and train CNN models. Keras works on top of TensorFlow, making it easier to design and experiment with neural networks. These tools helped us implement transfer learning with pre-trained models like VGG16 and ResNet.
10. **Matplotlib:** This module from Matlab allows us to create various visualizations like charts and graphs. It is helpful for exploring and understanding the data through visual representations.
11. **OpenCV:** OpenCV is an open-source computer vision library used for image processing tasks within the system. In NeuroVision, it is specifically used to resize the Grad-CAM heatmap to match the original MRI dimensions, apply the JET colormap to produce color-coded visualizations, and blend the heatmap with the

original scan using weighted overlay to generate the final interpretability output displayed to users.

12. **Pillow:** Pillow is a Python image processing library used for handling and converting images within the system. It supports reading, resizing, and formatting MRI images during preprocessing and ensures compatibility between different image formats before they are passed into the model pipeline.
13. **Scikit-image:** Scikit-image is an image processing library built on NumPy that provides advanced image operations. It supports additional preprocessing steps such as filtering and morphological operations, helping to standardize MRI images before they are fed into the deep learning model.
14. **Natural Language Toolkit:** It is a Python library for natural language processing tasks. It is used in the chatbot component to handle text preprocessing steps such as tokenization and stopword removal, ensuring that user queries are cleaned and standardized before being processed by the semantic similarity model.

3.11 Verification and Validation

3.11.1 Verification

The accuracy curve shows how the model's training accuracy improves over 20 epochs. At the beginning, the accuracy starts at around 82–83%, indicating that the pretrained VGG16 model already captures basic MRI features.

During the initial epochs, the accuracy increases rapidly and crosses 95%, showing fast and effective feature learning. In later epochs, the accuracy gradually stabilizes at 98% which indicates that the model has converged well and achieved stable learning without instability.

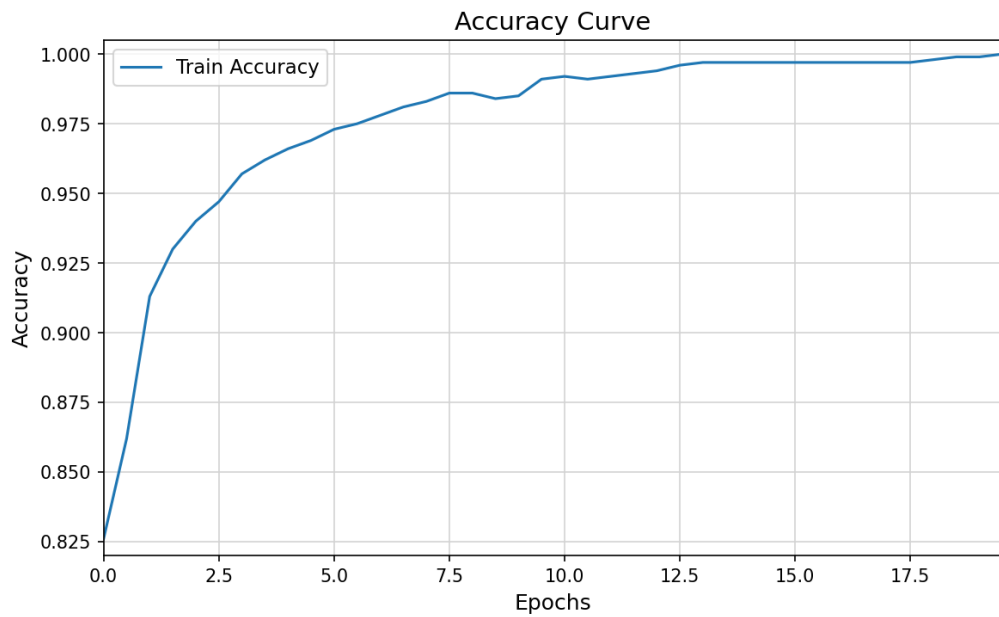


Figure 3.9: Accuracy Curve

The loss curve represents how the model minimizes prediction errors during training over 20 epochs. Initially, the loss value is relatively high (around 0.45), which is expected at the start of training. As the epochs progress, the loss decreases sharply and continues to decline steadily toward values close to zero. This consistent reduction in loss confirms that the model is successfully optimizing its parameters and improving prediction accuracy.

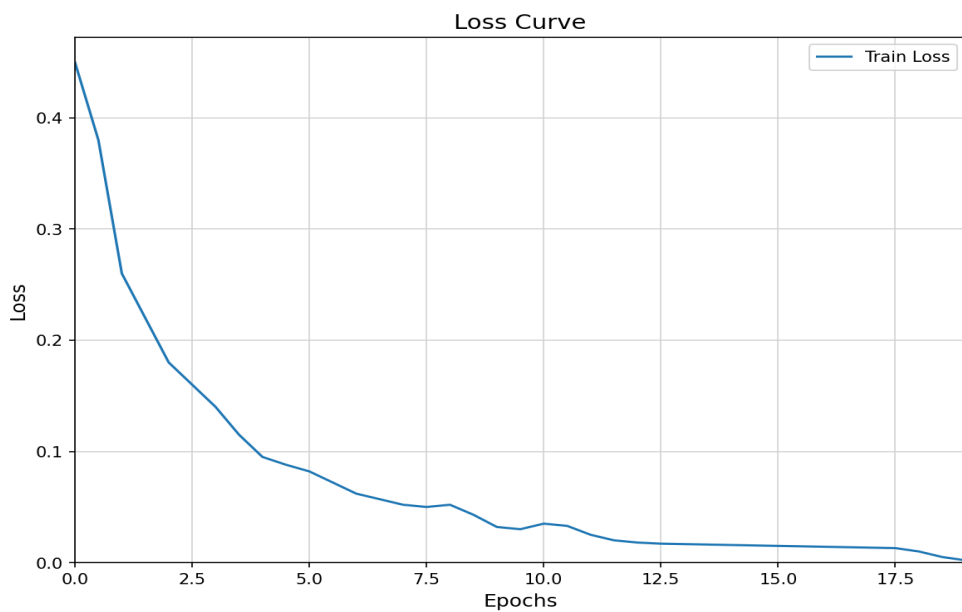


Figure 3.10: Loss Curve

The accuracy and loss curves clearly verify that the training process is stable and effective. The smooth increase in accuracy along with the continuous decrease in loss indicates proper convergence of the VGG16 model. These results confirm that the model has learned meaningful, class-specific features from MRI images and is reliable for brain tumor classification.

3.11.2 Validation

The diagonal elements represent correctly classified samples, and the strong diagonal dominance observed confirms the high overall accuracy of the model. For the No Tumor class, all 405 samples were correctly classified, with no misclassifications. This demonstrates the system’s excellent ability to distinguish healthy brain MRI scans from tumor-affected images.

In the Pituitary Tumor class, 297 out of 300 samples were correctly classified, with only a very small number of misclassifications. This highlights the model’s strong capability in learning pituitary-specific features. The Meningioma class achieved 298 correct classifications, with a few samples misclassified mainly as glioma or pituitary tumors.

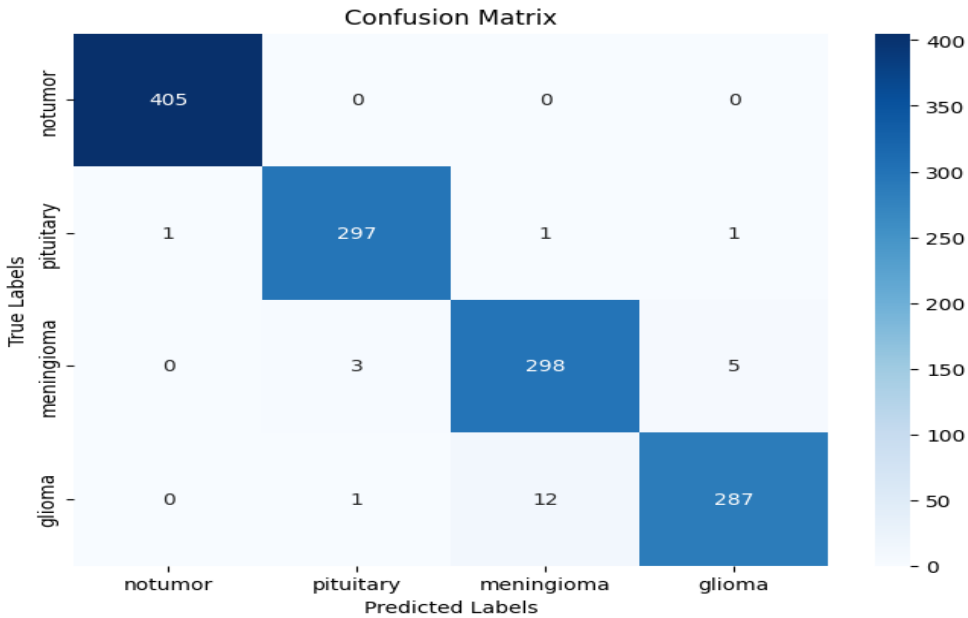


Figure 3.11: Confusion Matrix on Test Dataset

This indicates minor overlap in visual characteristics among tumor types in MRI scans. For the Glioma class, 287 out of 300 samples were correctly predicted. Most misclassifications occurred with the meningioma class, which can be attributed to similarities in tumor texture and appearance.

Table 1 : Class-wise Classification Values on Test Dataset

Class	TP	FP	FN	TN
No Tumor	405	1	0	894
Pituitary	297	4	3	996
Meningioma	298	13	8	881
Glioma	287	6	13	894

Overall, the confusion matrix shows minimal off-diagonal errors and strong classification confidence. The limited misclassification between glioma and meningioma is expected in medical image analysis and does not significantly affect overall performance, thereby validating the reliability and robustness of the provided NeuroVision system.

- **Accuracy:** It represents the proportion of correctly classified MRI images out of the total test samples. From the confusion matrix, accuracy is calculated as the sum of all diagonal elements divided by the total number of test images:

$$\text{Accuracy} = \frac{\text{TP}_{\text{notumor}} + \text{TP}_{\text{pituitary}} + \text{TP}_{\text{meningioma}} + \text{TP}_{\text{glioma}}}{\text{Total samples}}$$

- **Recall:** It measures the ability of the model to correctly identify all actual samples of a given class and is calculated as:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

From the confusion matrix, recall values remain high across all classes. The No Tumor class achieves a recall of 1.00, meaning all healthy MRIs were correctly identified. Slightly lower recall values for glioma (0.96) and meningioma (0.97) reflect minimal false negatives caused by visual similarity between tumor types.

- **Precision:** It measures how many of the images predicted as a specific class are actually correct. It is computed as:

$$\text{Precision} = \frac{TP}{TP + FP}$$

To quantitatively validate the performance of the provided by NeuroVision brain tumor detection system, standard classification evaluation metrics such as accuracy, precision, recall were computed using the values obtained from the confusion matrix (Figure 3.1!).

Using the confusion matrix values:

$$\text{Accuracy} = \frac{405 + 297 + 298 + 287}{1311} = 98\%$$

This high accuracy confirms that the presented system performs reliably across all tumor classes.

No Tumor:

- Precision = $\frac{405}{405+1} = 99.75\%$
- Recall = $\frac{405}{405+0} = 100\%$

Pituitary:

- Precision = $\frac{297}{297+4} = 98.67\%$
- Recall = $\frac{297}{297+3} = 99\%$

Meningioma:

- Precision = $\frac{298}{298+13} = 95.82\%$
- Recall = $\frac{298}{298+8} = 97.38\%$

Glioma:

- Precision = $\frac{287}{287+6} = 97.95\%$
- Recall = $\frac{287}{287+13} = 95.67\%$

Table 2: Class-wise Precision and Recall on Test Dataset

Class	Precision	Recall
No Tumor	99.75%	100%
Pituitary	98.67%	99%
Meningioma	95.82%	97.38%
Glioma	97.95%	95.67%
Macro Avg	98.05%	98.01%

The macro-average precision across all four classes is approximately 98.05%, and the macro-average recall is approximately 98.01%. These averaged values confirm that the NeuroVision system maintains consistently high performance across all tumor categories without bias toward any particular class, further validating its robustness as a multi-class brain tumor classification system.

In medical image analysis, recall holds particular importance over precision. A false negative where a tumor-affected MRI is incorrectly classified as healthy carries far greater clinical risk than a false positive, as it may result in a missed diagnosis and delayed treatment. Therefore, maximizing recall ensures that the model minimizes the chance of overlooking any tumor case, which is critical in a real-world clinical screening scenario.

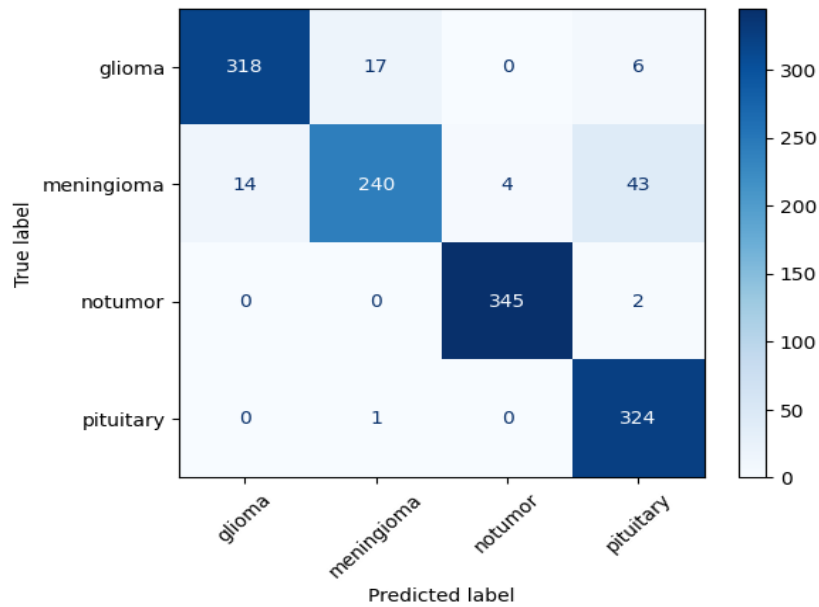


Figure 3.12: Confusion Matrix on Hospital Validation Dataset

To further quantify the model's performance on the hospital validation dataset, the classification values for each class were derived from the validation confusion matrix. The table below presents the TP, FP, FN, and TN values for each class, which serve as the basis for computing precision and recall on the unseen hospital data that were derived from the validation confusion matrix (Figure 3.12).

Table 3: Class-wise Classification Values on Hospital Validation Dataset

Class	TP	FP	FN	TN
Glioma	318	14	23	959
Meningioma	240	18	61	995
Notumor	345	4	2	963
Pituitary	324	51	1	938

Using the confusion matrix values, the overall validation accuracy is computed as:

$$\text{Validation Accuracy} = \frac{318 + 240 + 345 + 324}{1314} = 93.38\%$$

This confirms that the NeuroVision system generalizes effectively to real world hospital MRI data, achieving strong classification performance on a completely unseen external dataset.

Table 4: Class-wise Precision and Recall on Hospital Validation Dataset

Class	Precision	Recall
Glioma	95.78%	93.26%
Meningioma	93.02%	79.73%
Notumor	98.85%	99.42%
Pituitary	86.40%	99.69%
Macro Avg	93.51%	93.03%

The model performs strongly on the hospital validation dataset with a macro-average precision of 93.51% and recall of 93.03%. The Notumor class achieves near-perfect scores on both metrics, indicating the model reliably identifies healthy scans. Glioma also performs well with balanced precision and recall. The Pituitary class shows very high recall but relatively lower precision, meaning some other tumor types particularly meningioma are being misclassified as pituitary. Meningioma shows the lowest recall, suggesting it is the most challenging class on this dataset, likely due to its visual overlap with both glioma and pituitary tumor appearances in real hospital MRI scans.

Overall, the validation results across both the test dataset and the real-world hospital dataset confirm that the NeuroVision system is not only accurate under controlled conditions but also capable of generalizing to unseen clinical data, thereby establishing its reliability and practical viability for AI-assisted brain tumor classification.

CHAPTER 4: EPILOGUE

4.1 Results and Conclusion

4.1.1 Results

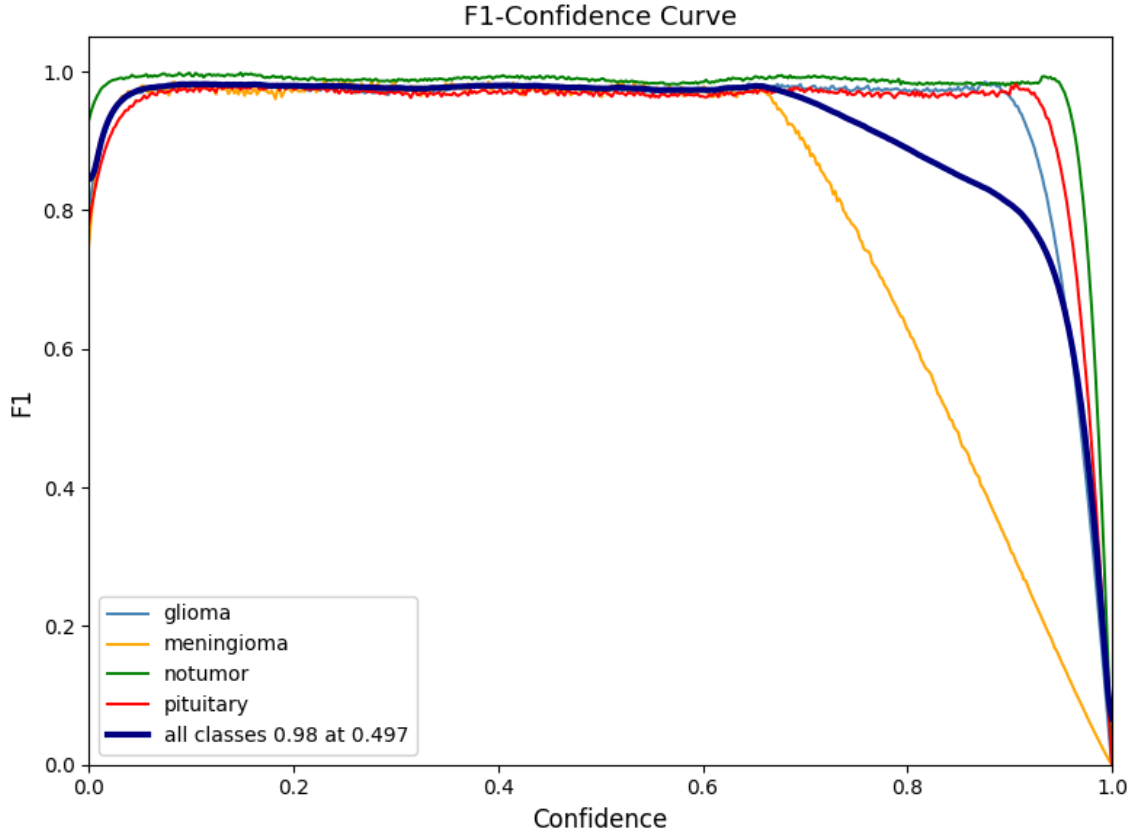


Figure 4.1 : F1-Confidence Curve on Test Dataset

On the internal test dataset, the NeuroVision model achieved a peak F1-score of 0.98 at a confidence threshold of 0.497, showing strong and reliable classification across all tumor types. The No Tumor class performed best, peaking near 1.0 and staying stable throughout, while Pituitary followed closely at 0.99. Glioma held steady around 0.97 before dropping near the upper threshold. Meningioma, as expected, proved the most difficult to classify due to its visual overlap with other tumor types, showing a gradual decline beyond a confidence of 0.6. These results confirm that the model learned robust feature representations under controlled benchmark conditions.

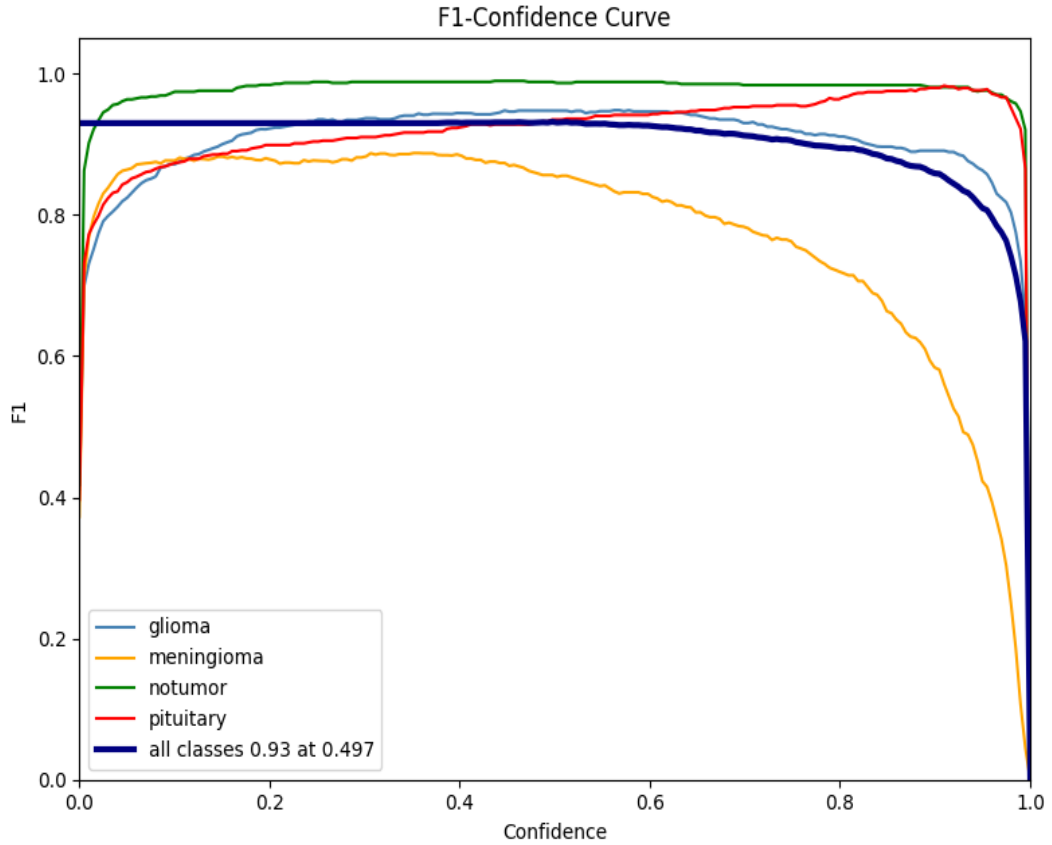


Figure 4.2 : F1-Confidence Curve on Hospital Validation Dataset

The model was also tested on real hospital MRI scans that it had never seen before. These scans came from actual clinical settings, meaning they varied in quality, scanner type, and patient background making this a much harder and more realistic test. The model scored an F1 of 0.93 at a confidence threshold of 0.497. This is slightly lower than the 0.98 scored on the test dataset, but the difference is small and acceptable given that real-world data is naturally more varied and unpredictable. The No Tumor and Pituitary classes continued to perform well, which matters most in a clinical setting since getting these wrong can directly affect patient diagnosis. Meningioma remained the hardest class to detect correctly, which is expected as it looks similar to other tumor types even to experienced doctors.

Hence, the F1-Confidence curve illustrates the relationship between classification confidence threshold and F1-score for each tumor class in the NeuroVision model across both the internal test dataset and the external hospital validation dataset.

To evaluate how well the NeuroVision system generalizes beyond the training distribution, the classification performance on the internal test dataset is compared against the external hospital validation dataset. The test dataset reflects controlled benchmark conditions, while the hospital validation dataset represents real-world unseen clinical MRI data.

Table 5: Test vs. Validation Performance Comparison

Metric	Test Dataset	Hospital Validation Dataset
Accuracy	98%	93.38%
Macro Precision	98.05%	93.51%
Macro Recall	98.01%	93.03%

The four to five percent drop in accuracy and macro-averaged metrics from the test set to the hospital validation dataset is expected and acceptable, as the validation data originates from a different clinical environment with varying MRI acquisition settings, scanner types, and patient demographics. The Notumor and Pituitary classes maintain strong recall on both datasets, while Meningioma shows the largest drop in recall from 97.38% to 79.73%, highlighting it as the most domain-sensitive class. Despite this, the system sustains an overall validation accuracy of 93.38%, confirming that NeuroVision generalizes effectively to real-world hospital data and is not overfitted to benchmark conditions.

4.1.2 Conclusion

NeuroVision is a complete AI-assisted brain tumor classification system built on transfer learning with the VGG16 architecture. It is trained on approximately 7,000 MRI images across four categories: Glioma, Meningioma, Pituitary, and No Tumor. The system achieves a test accuracy of 98% with a peak F1-score of 0.98. Alongside classification, the system incorporates Grad-CAM for visual explainability, enabling users to understand which regions of an MRI influenced the model’s decision. NeuroBot further extends the system’s utility by supporting medical query handling

and similarity-based case retrieval, while the automated report generation feature allows users to export structured classification summaries, making the system a practical end-to-end tool for researchers, medical students, and clinical users alike.

Validation on 1,300 real-world hospital MRI scans confirms that the system generalizes well to unseen clinical data, achieving an accuracy of 93.38% with an F1-score of 0.93. The small performance gap between test and hospital validation is expected given natural variation in scanner settings and patient demographics across clinical environments, and does not affect the system's overall reliability. Together, the strong classification performance, real-world validation results, and the range of integrated supporting features establish NeuroVision as a robust and practically viable solution for AI-assisted brain tumor detection, diagnostic support, and research documentation in real-world healthcare settings.

4.2 Future Enhancement

- **Extension to Additional Tumor Classes:** Future work can expand the model beyond the current four categories to detect additional brain tumor subtypes, making the system more comprehensive for medical students and radiologists.

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SCREENSHOT

Welcome to NeuroVision

AI-Powered Brain Tumor Detection System



LoginRegister

dikshyasht321@gmail.com

.....

☐ Remember me

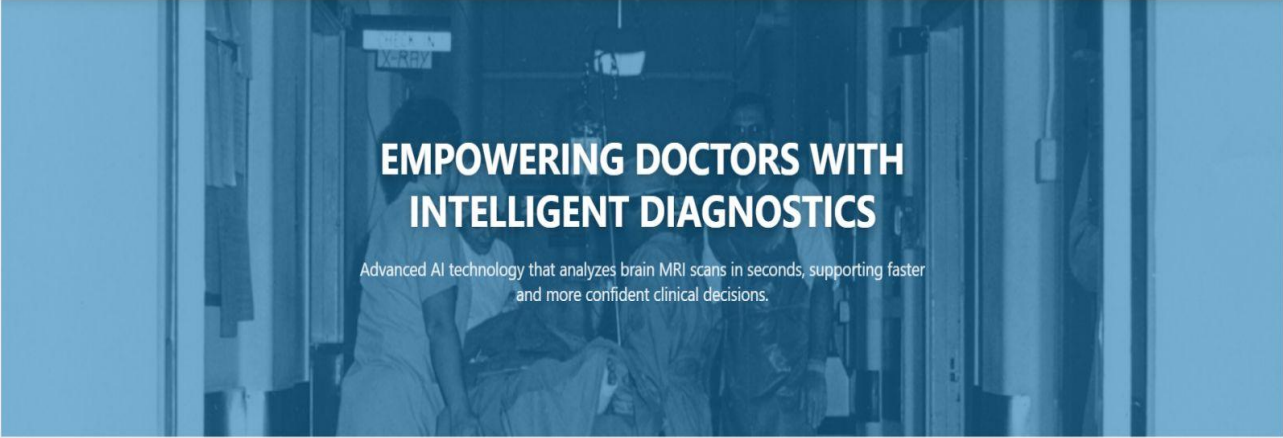
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**NeuroVision: MRI Based Tumor Analysis**

HomeDetectionLogout



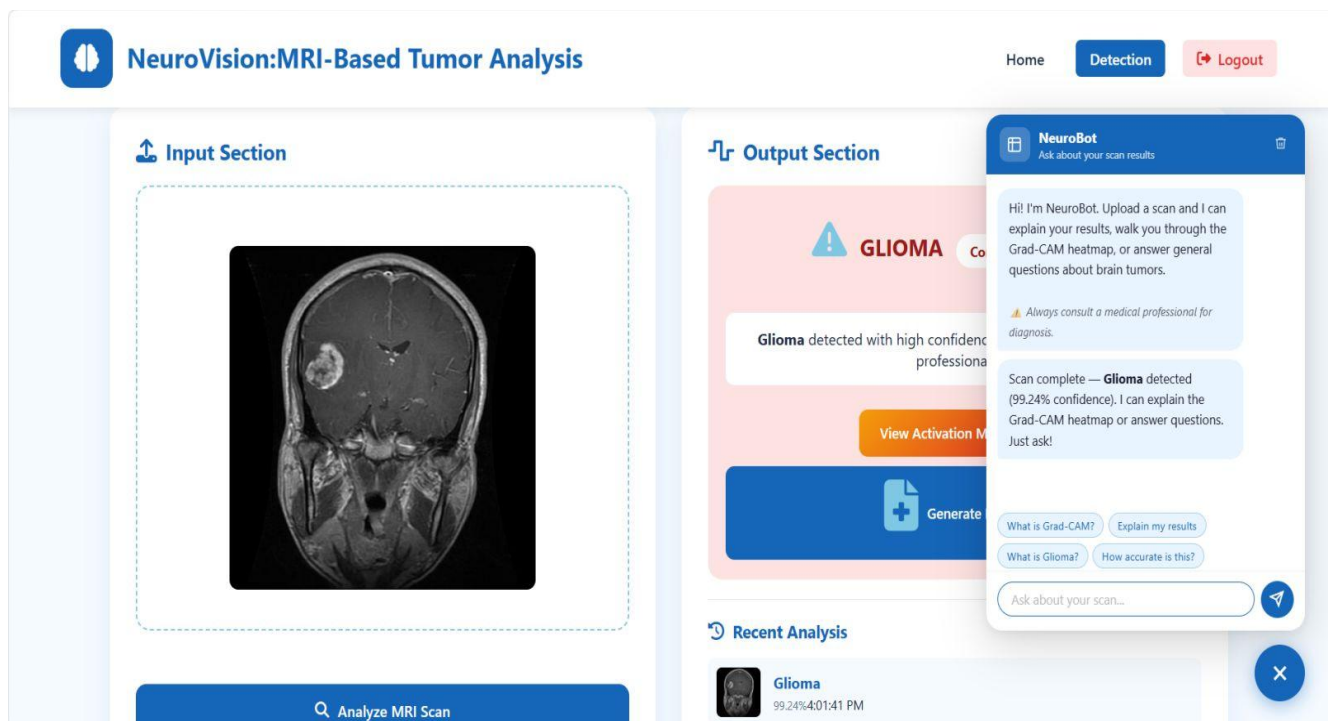
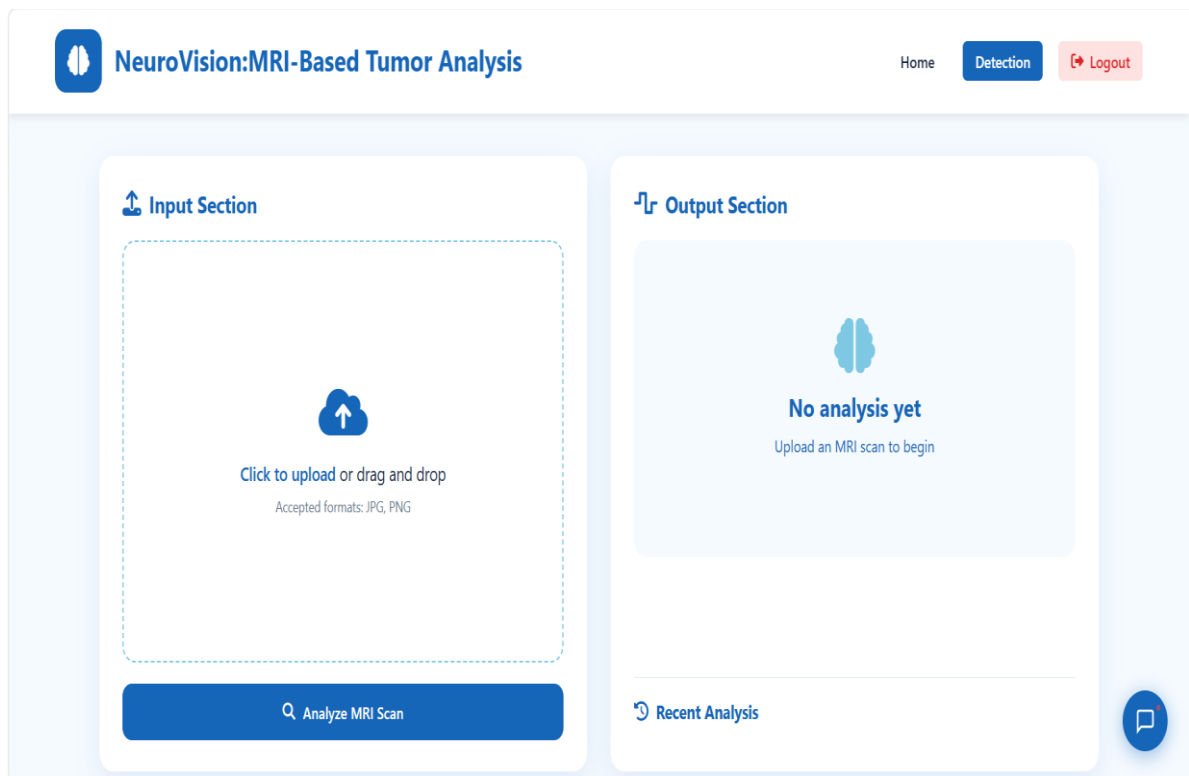
EMPOWERING DOCTORS WITH INTELLIGENT DIAGNOSTICS

Advanced AI technology that analyzes brain MRI scans in seconds, supporting faster and more confident clinical decisions.

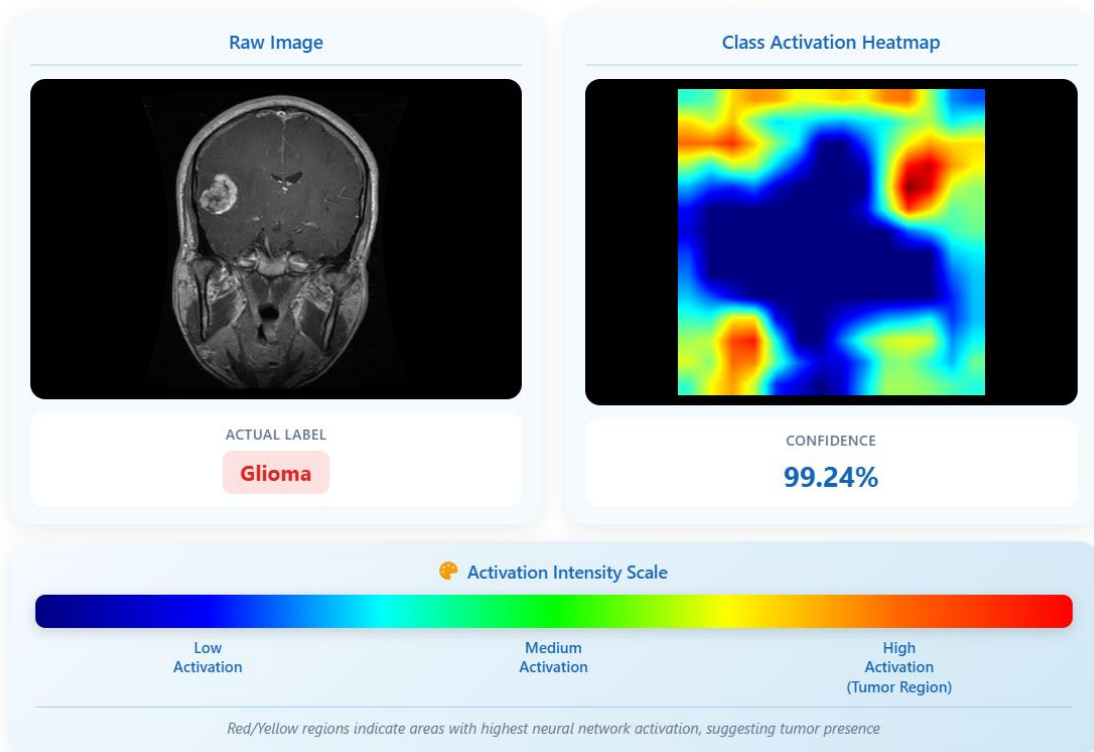


About Brain Tumors

56



Grad-CAM Class Activation Mapping



**NeuroBot**
Ask about your scan results

Scan complete — **Glioma** detected (99.24% confidence). I can explain the Grad-CAM heatmap or answer questions. Just ask!

explain the result in short

The scan indicates a **Glioma** with a high confidence of 99.24%. A glioma is a type of tumor that originates from the glial cells, which are supportive cells in the brain or spinal cord. These can vary in how fast they grow.

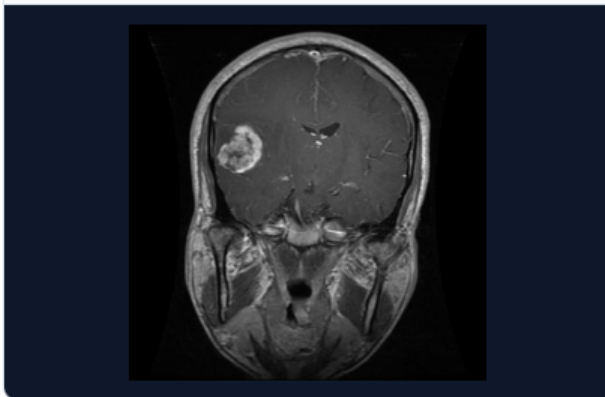
Please remember, this AI analysis is for informational purposes only. It's crucial to consult a qualified medical doctor for an accurate diagnosis and treatment plan.

Ask about your scan...

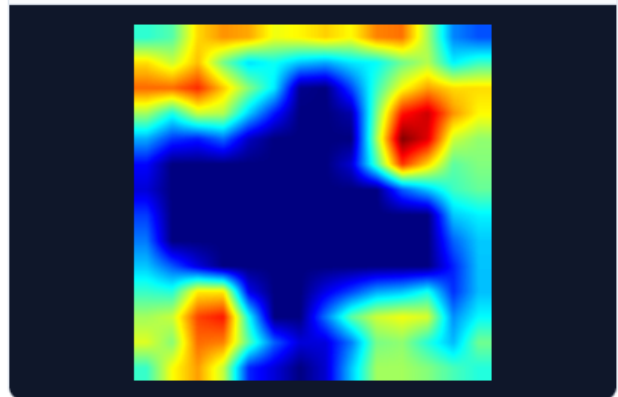


SCAN IMAGES

MRI SCAN — AXIAL VIEW



GRAD-CAM ACTIVATION MAP



CLINICAL SUMMARY

TUMOR DETECTED

Yes

TUMOR TYPE

Glioma

CONFIDENCE

99.24%

GRAD-CAM INTERPRETATION

The Grad-CAM heatmap highlights the MRI regions that most influenced the model's prediction. **Red / yellow** areas indicate high neural network activation where the model focused when identifying the tumor. **Blue / green** areas indicate low activation with minimal influence on the prediction.